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AN ELEMENTARY INTRODUCTION TO THE CONSTRUCTION AND THE ANALYSIS OF PERFECTLY MATCHED LAYERS FOR TIME DOMAIN WAVE PROPAGATION

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As soon as people have been interested in the numerical simulation of wave propagation phenomena, particularly in the time domain, they had to face the delicate question of realizing "transparent" artificial boundaries. Indeed, many problems of this type are posed in a unbounded domain (or in a domain which is very large with respect to a zone of interest in which one wishes to observe the solution) : one thus has to bound artificially the computational domain in such a way that the non physical boundaries are "transparent" for the solution. Due to the crucial importance of this subject, but also because of its difficulty, this question has aroused the curiosity from many researchers, first from engineers, next from applied mathematicians. It would be much too ambitious to give a complete panorama of a research that began almost forty years ago (see however the article by T. Hagstrm [20] which constitutes a quite complete review of the domain) but one can now measure how much the many ingenious ideas proposed by the one or the others have permitted to accomplish impressive progress in this matter. Nowadays, if one restricts oneself to linear wave propagation problems, one can consider that, even though some questions remain open, satisfactory solutions are available for a large number of applications, in particular in the "canonical case" I will have in mind throughout this paper, namely the case where one wishes to reduce numerical computations to a convex bounded domain, assuming that the exterior domain is homogeneous and source free.

Recently, in particular for the applications in acoustics and electromagnetism, important progress has been achieved in the use of "exact" boundary conditions associated to homogeneous exterior domains. One usually introduces a boundary operator called "Dirichlet to Neumann" (DtN) operator that can be represented by using separation of variables in the exterior domain (in 3D, one will use typically spherical harmonics [19], which requires a spherical artificial boundary) or through the coupling with an integral representation formula of the exterior solution (which relies on the explicit knowledge of the Green's function of the problem) [4].

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However, these techniques, beyond the fact that they remain restricted to a relatively limited field of applications, have the disadvantage to be nonlocal in space and time : they remain quite difficult to implement and rather costly in terms of computational time and storage, despite of the recent progress accomplished is fast algorithms for the evaluation of such nonlocal operators, such as the multipole method [18]. That is why, in many computational codes, one prefers to use solutions involving local operators, typically differential operators, which lead to a much simpler implementation. Such solutions, except in 1D, can not provide transparent boundary conditions : one speaks of "approximate" or "absorbing" boundary conditions. Of course, these conditions involve approximation parameters (devoted to tend to 0 or $+\infty$) and will provide the true solution only when "passing to the limit". In this class of methods, two large families of methods can be opposed:

- Absorbing boundary conditions (ABC's): The idea is to complete the interior mathematical model (namely the interior PDE's) with one or several boundary conditions that can be seen as partial differential equations along the boundary. Such conditions are supposed to minimize the spurious reflections that they generate. The pioneering works of Engquist and Majda [15], [16] on the wave equation (in the half space, i. e. with a flat artificial boundary) are in this spirit. They proposed a hierarchy of boundary conditions that are classified with an integer N, namely the order of the boundary condition that I will not precisely define here. Let us simply explain that the greater is N, the higher is the order of the differential operators (which means that the boundary condition is more and more costly in terms of computational time) and the better is the accuracy of the approximate solution, as it has been recently proven in [14] for instance. These initial works have led to an intensive research in applied mathematics aiming at extending this type of techniques to other equations, the treatment of curved boundaries, the treatment of corners,...

- Absorbing layers : the idea here is that the absorption of the waves that "want" to leave the computational domain is not achieved through a boundary condition but inside a (hopefully small) layer surrounding this computational domain. This layer will be made of an "absorbing material" whose behavior remains described with the help of partial differential equations and inside which the waves are damped (hopefully exponentially fast) during their propagation. In such a case, the boundary condition that has to be prescribed at the external boundary of the absorbing layer has no particular importance as soon as it leads to a well posed problem...

The idea, physically rather natural, of absorbing layers has first been introduced by engineers at the beginning of the 70's. The idea was to join the physical non absorbing domain to an absorbing medium of physical nature (typically, in electromagnetism one will use a conductive material, in mechanics a visco-elastic material, ...). This approach has been progressively abandoned - during the 80's - at the benefit of absorbing boundary conditions for various reasons, in particular because it was not supported - contrary to ABC's - by a solid and precise theoretical background but also because its practical use required what I will cal some "engineer intuition". In the mid 90's, the

works of J. P. Bérenger [8], [9], who, for Maxwell's equations, invented the concept of Perfectly Matched Layers (PMLs), have revolutionized the domain and replaced the methods of absorbing layers as a very attractive and powerful alternative to absorbing boundary conditions. The success of PMLs, whose principle is based on a simple but very clever idea, can be explained by several good reasons among which

- the fact that PML's are particularly simple to implement (in particular by comparison with higher order absorbing boundary conditions),
- the fact that they provide in many applications remarkable performances (even when the theory does not completely predict it),
- the fact that they can be applied without extreme additional complexity to a great number of equations (including acoustics, electromagnetism, elastodynamics,...)

However, to temperate any excessive enthusiasm, I will add that

- although substantial progress have been accomplished during the past ten years (which I will try to show later) the mathematical analysis of these methods is still not sufficiently well understood (at least from my point of view).
- there exist physical models for which they do not work because they lead to unstable mathematical models.
- the winner of the competition PMLs agains ABCs is not clearly designated yet and a fair comparison remains to be done.

As many (really many) things have already been written on PML's, I will choose to insist in this text on a particular point of view, which I hope to be well adapted to a reader who is an applied mathematician. More precisely, I will pursue a double goal:

- Give a very introductory exposition of the construction of PML's, which is not the one originally the one of Bérenger but better corresponds to my understanding of the method,
- Give a kind of state of the art in the matter of mathematical results about PML's, with a particular emphasis on the works done by my collaborators and myself at INRIA.

In this spirit, I have chosen to organize the content as follows. Chapter 1 concerns the 1D case and has essentially a pedagogical motivation. After a brief introduction on classical absorbing layers, I will present - starting from the classical concept of impedance adaptation - the construction of PML's for the 1D wave equation. In this very simple case, everything becomes quite simple and can be analyzed "by hand". Chapter 2 is devoted to Bérenger's PML's for a general hyperbolic system in dimension 2 (once one has understood the two-dimensional case, the extension to higher space dimensions does not pose any problem). I will show how, from a very mathematical and formal idea - the use of a "complex" (in the sense of imaginary numbers) change of variable - one can derive PML's that generalize the ones proposed by Bérenger for Maxwell's equations. In a second step, I will restrict myself to the 2D acoustic wave

equation on which I will explain in detail the absorption mechanism for plane waves, the treatment of corners, the interest of a variable (in space) absorption coefficient,... The chapter 3, which concerns the mathematical analysis of PML models, will report more of some of my recent research. After having recalled the concepts of well-posedness and stability, I will present some results from the literature. These results are incomplete. Some of them enlighten some instability for certain classes of equations which are however of real interest for applications. Finally, some open problems will be mentionned.

1 Perfectly Matched Layers in dimension 1

Let us consider the 1D wave equation on the real line, with velocity 1:

$$\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} = 0, \quad x \in \mathbb{R}, \quad t > 0 \tag{1}$$

The Cauchy problem for this equation models in particular the small amplitude movements of a string submitted for instance to an initial deformation: such deformations propagate with velocity 1 without any modification of their shape or amplitude. In what follows, our objective is to characterize, for computational purpose, the restriction of the solution to the half-line $x < 0$ by introducing an artificial boundary at $x = 0$, assuming that the initial data are supported in this half-line. In this case, there is a very simple transparent condition, which is local (but this is no longer true in dimension > 1):

$$\frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = 0, \quad x = 0, \quad t > 0. \tag{2}$$

The (very academic) question that we address in this section is the following : how is it possible to realize a "quasi"-transparent condition via an absorbing layer occupying the interval $[0, L]$?

1.1 Classical absorbing layers

It is well known physically that one introduces some mechanisms of dissipation of energy, and thus wave damping, if one adds to the equation (1):

- a fluid friction term: $\quad + \nu \dfrac{\partial u}{\partial t}, \quad \nu \geq 0,$
- a viscous friction term: $\quad -\dfrac{\partial}{\partial x}\Big(\nu^* \dfrac{\partial^2 u}{\partial x \partial t}\Big), \quad \nu^* \geq 0,$

which leads to the equation (where ν and ν^* may vary in x):

$$\frac{\partial^2 u}{\partial t^2} + \nu \frac{\partial u}{\partial t} - \frac{\partial^2 u}{\partial x^2} - \frac{\partial}{\partial x}\Big(\nu^* \frac{\partial^2 u}{\partial x \partial t}\Big) = 0.$$

The most visible effect (see figure 1, left picture) is the (exponential) decay of the amplitude of the wave during its propagation, a decay which increases when the

absorption coefficients ν and ν^* increase. The second one is wave dispersion which affects the propagation velocity of the waves. Analytically, looking for time oscillating solutions of the form:

$$u(x) \exp i\omega t, \quad \omega > 0, \quad \text{(given)} \tag{3}$$

leads, assuming that ν et ν^* are constant, to:

$$u(x) = A \exp ik(\omega)x + B \exp -ik(\omega)x,$$

where $k(\omega)$ is the complex wave number characterized by:

$$k(\omega)^2 = \frac{\omega^2 + i\omega\nu}{1 + i\omega\nu^*}, \quad \mathcal{I}m\ k(\omega) \leq 0. \tag{4}$$

As $\mathcal{I}m\ k(\omega) < 0$ as soon ν or ν^* is nonzero, we have evanescent waves, namely solutions that decay exponentially in one direction. A natural idea for creating an (infinite in a first step) absorbing layer consists in taking:

$$\begin{cases} \nu(x) = 0, & \nu^*(x) = 0, & \text{for } x < 0, \\ \nu(x) = \nu > 0, & \nu^*(x) = \nu^* > 0, & \text{for } x \in [0, L], \end{cases}$$

and impose for example (this should be unessential) a homogeneous Neumann condition at $x = L$. In practice, one is faced to the following dilemma that we illustrate on figure 1 (center and right pictures):

- When ν et ν^* increase, one improves the absorption phenomenon, which should a priori allow to diminish the length L of the layer.
- On the other hand, the two media being more and more different, the amplitude of the reflected wave increases and very few energy penetrates the absorbing layer.

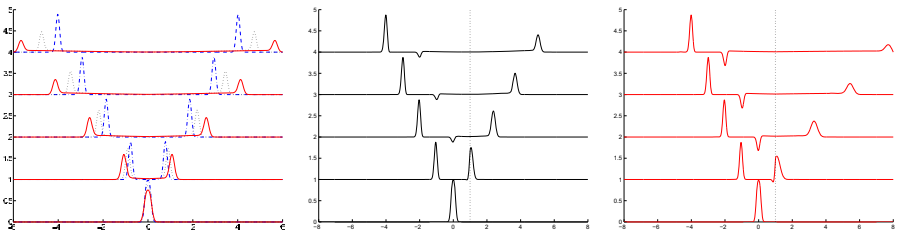


Figure 1: Left: Propagation in a "visco-elastic" (In blue : no absorption, in black weak absorption, in red strong absorption). Right and center: reflection by a visco-elastic layer. The absorption is stronger on the right. The horizontal axis corresponds to the space variables. One represents snapshots of the solution at equidistant instants.

As a consequence, in some sense, "the more absorbing is the medium, the less absorbing it is" ! By the way, this corresponds to a real technological difficulty met by physicists and engineers who want to build real absorbing coatings, for stealth technology for instance. This can be emphasized by the analysis. Let us consider the case of two semi-infinite media $x < 0$ and $x > 0$ with $\nu = \nu^* = 0$ for $x < 0$ but are strictly positive for $x > 0$. Let us consider particular solutions of the form (this is the reflection-transmission analysis of an incident harmonic wave):

$$\begin{cases} u(x) = \exp i\omega(t-x) + R(\omega)\ \exp i\omega(t+x), & \text{for } x < 0, \\ \\ u(x) = T(\omega)\ \exp -ik(\omega)x\ \exp i\omega t, & \text{for } x > 0. \end{cases}$$

where by definition

- $\exp i\omega(t-x)$ is the incident wave (of amplitude 1),
- $R(\omega)\exp i\omega(t+x)$ is the reflected wave and $R(\omega)$ is the relection coefficient,
- $T(\omega)\exp -ik(\omega)x\ \exp i\omega t$ is the transmitted wave and $T(\omega)$ is the transmission coefficient.

One obtains the expression of the coefficients $R(\omega)$ et $T(\omega)$ writing the transmission conditions at the interface $x = 0$. Introducing $k^*(\omega) = (1 + i\omega\nu^*)\ k(\omega), k(\omega)$ being given by (4), we get:

$$R(\omega) = \frac{k^*(\omega) - \omega}{k^*(\omega) - \omega}, \quad T(\omega) = 1 + R(\omega).$$

As $\lim_{\nu \to +\infty} |k^*(\omega)| = \lim_{\nu^* \to +\infty} |k^*(\omega)| = +\infty$, we deduce that

$$\lim_{\nu \to +\infty} |R(\omega)| = \lim_{\nu^* \to +\infty} |R(\omega)| = 1.$$

To overcome this phenomenon, an intuitive idea consists in increasing progressively ν et ν^* inside the layer, in order to dilute the reflections et permit that the wave penetrates better in the absorbing layer. This type of strategy has been studied for instance in [23]. However the choice of good profiles $\mu(x)$ et $\nu(x)$ as well as the determination of L are not completely clear and that is why the practical and robust use of such layers remains quite delicate.

1.2 The Perfectly Matched Layers in 1D.

The main drawback of the layers studied in the previous section is the reflection of waves at the interface between the real medium and the absorbing one. This raises the natural question: can one conceive an absorbing layer which would be "perfectly matched", that is to say that would generate no reflection at the interface with the physical medium ? If we restrict ourselves to the type of absorbing layer considered in the previous paragraph, the answer is no, even if ν et ν^* are authorized to vary in x. However, we shall see that the answer becomes positive if one authorizes non physical absorption mechanisms.

Construction of PML's by adaptation of impedance. As a starting point, let us recall what is the adaptation of impedance. For this, let us come back to the wave equation with variable coefficients

$$\rho(x)\frac{\partial^2 u}{\partial t^2} - \frac{\partial}{\partial x}\Big(\mu(x)\frac{\partial u}{\partial x}\Big) = 0, \tag{5}$$

where coefficients ρ et μ are bounded and strictly positive functions. Let us introduce

- the propagation velocity $c(x) = \big(\mu(x)/\rho(x)\big)^{1/2}$,
- the impedance $z(x) = \big(\mu(x)\rho(x)\big)^{1/2}$.

Let us now consider a medium with two homogeneous layers:

$$\begin{cases} c(x) = c, \qquad z(x) = z, \qquad \text{pour } x < 0, \\ \\ c(x) = c_*, \qquad z(x) = z_*, \qquad \text{pour } x > 0. \end{cases}$$

One analyzes the reflection and transmission of a time harmonic incident wave in the medium $\{x < 0\}$ by studying the solutions of the form

$$\begin{cases} u(x) = \exp i(\omega t - kx) + R(\omega)\ \exp i(\omega t + kx), & \text{pour } x < 0, \quad (k = \omega/c) \\ \\ u(x) = T(\omega)\ \exp(\omega t - k_* x) & \text{pour } x > 0, \quad (k_* = \omega/c_*). \end{cases}$$

The traduction of the transmission conditions at $x = 0$ provides the expression of the reflection and transmission coefficients:

$$R = \frac{z - z_*}{z + z_*}, \quad T = \frac{2z}{z + z_*}\ .$$

In particular, one observes that if the two impedances z et z_* are equal (one says then that there is adaptation of impedance) the reflection coefficient vanishes. In other words, the medium $\{x > 0\}$ is transparent for the medium $\{x < 0\}$ - or equivalently perfectly matched. Nevertheless, the medium $\{x > 0\}$ being non absorbing, we simply have postponed the problem of the transparent condition. In fact, we shall overcome this difficulty by considering (unphysical) media in which "the velocity of waves may become complex" as in a visco-elastic medium !

Which follows is rather formal from the mathematical point of view but provides in my opinion a good intuitive understanding of the construction of PML's.

We first place ourselves in the frequency domain : in other words we look at solutions of the form (3). This is not restrictive since we treat a linear equation whose coefficients do not depend on time. The equation (5) becomes

$$-\rho(x)\ \omega^2 u - \frac{\partial}{\partial x}(\mu(x)\frac{\partial u}{\partial x}) = 0\ . \tag{6}$$

Let us now consider $d(\omega) \in \mathbb{C}$ and,

$$\begin{cases} \rho(x) = \rho, & \mu(x) = \mu, & \text{for } x < 0, \\ \rho(x) = \dfrac{\rho}{d(\omega)}, & \mu(x) = \mu\, d(\omega), & \text{for } x > 0, \end{cases}$$

By construction, the product $\rho(x)\mu(x)$ is constant; in other words, the impedance in the medium $\{x > 0\}$ in the same than the impedance in the medium $\{x < 0\}$ ($z = \sqrt{\rho\mu}$). We thus have adaptation of impedance. However, the velocity $c = \sqrt{\mu/\rho}$ becomes complex : it is equal to $c\, d(\omega)$, which allows us to make the medium $\{x > 0\}$ absorbing if one chooses $d(\omega)$ adequately. To build this function, we have to keep in mind that, for the applications to time domain, coming back to the time domain should lead to differential operators in time. This means to impose that :

$$d(\omega) \textit{ is a rational fraction with real coefficients in the variable } i\omega. \tag{7}$$

In the medium $\{x > 0\}$, harmonic solutions are of the form (I reincorporate on purpose the time dependency)

$$\exp i\big(\omega t - k^*(\omega)\, x\big), \quad k^*(\omega) = \pm \frac{\omega}{c}\, d(\omega)^{-1}. \tag{8}$$

Let us set $d^{-1} = a + ib, (a, b) \in \mathbb{R}^2$. We impose to the real part a to be positive. It is the presence of the imaginary part b which will govern the absorption of waves. Indeed

$$k^*(\omega) = k_R(\omega) + ik_I(\omega), \quad k_R(\omega) = a\, \frac{\omega}{c}, \quad k_R(\omega) = b\, \frac{\omega}{c},$$

and the solution (8) can be rewritten:

$$\exp i\omega \left(t \pm \frac{ax}{c}\right)\ \exp\Big(\mp\ \omega\ \frac{bx}{c}\Big), \tag{9}$$

This corresponds to a wave which propagates with the velocity c/a in the direction $\pm\, x > 0$ et decays exponentially in the direction $\pm\, \omega bx > 0$ with a penetration length (or skin depth)

$$\ell(\omega) = |k_I(\omega)|^{-1} = \frac{c}{|\omega\ b|}\ .$$

At least intuitively, one realizes an absorbing medium if the amplitude decays in the same direction than the one in which it propagates (doing so, one absorbs the waves that propagate "to the right" as well as those that propagate "to the left") which imposes a priori:

$$\omega\ b < 0\ . \tag{10}$$

A simple choice, coherent with condition (7) consists in taking:

$$a = 1, \quad b = -\frac{\sigma}{\omega}, \tag{11}$$

where $\sigma > 0$ is called the absorption coefficient. With this choice, we observe in particular that:

- the absorption is uniform with respect to the frequency, the penetration length being :
$$\ell(\omega) = \frac{c}{\sigma} ,$$
- the propagation velocity of harmonic waves remains equal to c in the absorbing layer.

To reinterpret in time the propagation equation in the layer we remark that the choice (11) corresponds to
$$d(\omega) = \frac{i\omega}{i\omega + \sigma} . \tag{12}$$
The equation (6) in the medium $\{x > 0\}$ can thus be rewritten as
$$\rho\,(\sigma + i\omega)\,u - \frac{\partial}{\partial x}\left(\mu\,(\sigma + i\omega)^{-1}\,\frac{\partial u}{\partial x}\right) = 0, \tag{13}$$
which it is useful to rewrite as a first order system (introducing the new unknown v)
$$\begin{cases} \rho\,(\sigma + i\omega)\,u - \dfrac{\partial v}{\partial x} = 0, \\ \\ \mu^{-1}(\sigma + i\omega)\,v - \dfrac{\partial u}{\partial x} = 0, \end{cases}$$
which corresponds 1D time domain PML system
$$\begin{cases} \rho\left(\dfrac{\partial u}{\partial t} + \sigma u\right) - \dfrac{\partial v}{\partial x} = 0, \\ \\ \mu^{-1}\Big(\dfrac{\partial v}{\partial t} + \sigma v\Big) - \dfrac{\partial u}{\partial x} = 0. \end{cases} \tag{14}$$

Remark 1 *If one eliminates v in (14), one gets the equation (in the case $\rho = \mu = 1$)*
$$\frac{\partial^2 u}{\partial t^2} + 2\sigma\frac{\partial u}{\partial t} + \sigma^2 u - c^2\frac{\partial^2 u}{\partial x^2} = 0. \tag{15}$$
By comparison with the classical layers presented in the previous section, we see that we do not have to add a viscous friction term but that we need (at least this is sufficient) to add the purely dispersive 0 order term $\sigma^2 u$.

To understand correctly how this equation is coupled to the equation (1) through the interface $x = 0$, we need to make precise the transmission conditions, which are somewhat non classical :
$$\begin{cases} u(0^-, t) = u(0^+, t), \\ \\ \left(\dfrac{\partial^2 u}{\partial x \partial t} + \sigma\dfrac{\partial u}{\partial x}\right)(0^-, t) = \dfrac{\partial^2 u}{\partial x \partial t}(0^+, t) . \end{cases} \tag{16}$$
These conditions correspond in fact to the continuity of u and v in the first order formulation.

Remark 2 *The choice (11) (or (12)) is the one that gives the original PMLs initially proposed by Bérenger. It is one choice among many other possible ones, compatible with the conditions (7) and (10) (there is thus a lot of room for the construction of "new PML's"...), for instance:*

$$d(\omega) = 1 + i\omega\sigma, \quad \sigma > 0.$$

It is easy to see that the resulting PML equation is also dissipative, remains of "hyperbolic nature" (one still has propagation with finite velocity) and yields an exponential absorption of harmonic waves. However, this absorption is no longer uniform : the penetration length of plane waves depends on the frequency.

Construction of PML's through a complex change of variable. One can recover the system (14) by another method whose physical meaning is not obvious but which is more directly generalizable to higher space dimensions. First we remark that, at least when the coefficients ρ et μ are constant, the general solution of the Helmholtz equation (6), namely a linear combination of two exponentials, can be extended analytically to the complex plane (with respect to the space variable x). This extension still satisfies (6) in $\mathbb{C}$ (the differentiation being understood in the sense of functions of the complex variable). In some sense, one can imagine the existence of a "super-reality" in which the space variable X (the notation I use for the "extended" space variable) would vary in the complex plane and whose the "physical world" that we observe every day would only be the "trace" on the real axis. An idea, which is then natural, is to observe the solution along a new path ($\neq \mathbb{R}$) in the complex plane, such that

(i) this path can be easily parameterized with the true physical variable x $\big(\, X = X(x) \,\big)$, in such a way that the new "computational domain" can be identified to the physical one,

(ii) this path coincides with the real axis for $x < 0$, so that the observed solution coincides with the physical one for $x < 0$ (a way to ensure that the new medium is perfectly matched),

(iii) the solution decays exponentially along this path when x describes the half-line $x > 0$ by increasing values, so that an absorbing medium is realized,

(iv) After come-back to the "time domain", the variations of the "new solution" can be described with partial differential equations (this is criterion (7)).

One realizes that all these criteria are satisfies with the change of variable :

$$X(x) = x + \frac{1}{i\omega} \int_0^x \sigma(\xi)\, d\xi,$$

where $\sigma(x)$ is a damping function satisfying typically

$$\sigma(x) = 0 \text{ for } x < 0, \quad \sigma(x) > 0 \text{ for } x > 0.$$

Remarking that, if u and $\widetilde{u}$ are related by $\widetilde{u}(x) = u\big(X(x)\big)$ then

$$\frac{\partial \widetilde{u}}{\partial x}(x) = \frac{i\omega}{i\omega + \sigma(x)} \frac{\partial u}{\partial x}\big(X(x)\big),$$

one sees that if u is solution of (6), then $\widetilde{u}$ satisfies

$$-\frac{i\omega}{i\omega+\sigma}\frac{\partial}{\partial x}\Big(\frac{i\omega}{i\omega+\sigma}\,\mu\frac{\partial \widetilde{u}}{\partial x}\Big) - \rho\,\omega^2\widetilde{u} = 0\,.$$

that is to say (13). This approach shows that one still gets a "perfectly matched" layer even with a varying absorption profile. Moreover, if ρ and μ are constant, a basis of solutions of (13) is given by:

$$\exp i\omega x\ \exp\left(-\int_0^x \sigma(\xi)\,d\xi\right), \quad \exp -i\omega x\ \exp\left(+\int_0^x \sigma(\xi)\,d\xi\right).$$

Remark 3 *It is easy to see that :*

- *to satisfy the exponential decay criterion (iii), it suffices that the path $X(x)$ be included in the complex half-plane $\mathcal{I}m(X) > 0$,*
- *the criterion (iv) corresponds to the fact that the change of variable must depend on the $i\omega$ as a rational function with real coefficients.*

Analysis of 1D PML's. The well-posedness, even more the dissipative nature, of the problem (14) is trivial. It suffices for instance to remark that one has the energy identity

$$\frac{d}{dt}\left\{\frac{1}{2}\int\left(\rho\,|u|^2 + \mu^{-1}|v|^2\right)\,dx\right\} + \int \sigma\,\left(\rho\,|u|^2 + \mu^{-1}|v|^2\right)\,dx = 0\,. \tag{17}$$

The "perfect matching" property that we looked for can be expressed mathematically with the following result that, for simplicity, we state in the case $\mu = \rho = 1$:

Theorem 1 *Let (u, v) the solution of the Cauchy problem on the real line $\mathbb{R}$ associated to the system (the 1D wave equation written as a first order system)*

$$\frac{\partial u}{\partial t} - \frac{\partial v}{\partial x} = 0, \quad \frac{\partial v}{\partial t} - \frac{\partial u}{\partial x} = 0, \tag{18}$$

and initial data u_0 et v_0 with support in the half-space $x < 0$. Let (u^*, v^*) be the corresponding "PML solution", namely the solution of the Cauchy problem associated with the system (14) (with $\rho = \mu = 1$) for the same initial data. One then has the following relation

$$\begin{cases} u^*(x,t) = u(x,t)\ e^{-\int_0^x \sigma(\xi)\,d\xi}, \\ \\ v^*(x,t) = v(x,t)\ e^{-\int_0^x \sigma(\xi)\,d\xi}\,. \end{cases} \tag{19}$$

Proof :. The proof is elementary. The Cauchy problem for (14) being well posed, it suffices to check that the pair (u^*, v^*) given by formulas (19) satisfies the equations (14). As

$$\frac{\partial u}{\partial x} = \Big(\frac{\partial u^*}{\partial x} + \sigma\ u^*\Big)\ e^{\int_0^x \sigma(\xi)\ d\xi}, \quad \frac{\partial v}{\partial x} = \Big(\frac{\partial v^*}{\partial x} + \sigma\ v^*\Big)\ e^{\int_0^x \sigma(\xi)\ d\xi},$$

(u^*, v^*) satisfies the system of equations:

$$\frac{\partial u^*}{\partial t} - \frac{\partial v^*}{\partial x} - \sigma\ v^* = 0, \quad \frac{\partial v^*}{\partial t} - \frac{\partial u^*}{\partial x} - \sigma\ u^* = 0, \tag{20}$$

To conclude, it suffices to remark that (see below):

$$\sigma(u^* + v^*) = 0, \tag{21}$$

and to replace σu^* (respectively σv^*) by $-\sigma v^*$ (respectively $-\sigma u^*$) in the first (respectively second) equation of (20) to get (14).

To get (21), we remark that for $x < 0$, $\sigma = 0$ while for $x > 0$, $u^* = -v^*$. Indeed, from (18) we deduce in particular that:

$$(\frac{\partial}{\partial t} - \frac{\partial}{\partial x})\ [u+v] = 0 \quad \text{which yields} \quad [\,u+v\,]\ (x,t) = [\,u_0+v_0\,]\ (x+ct), \quad x > 0, \quad t > 0.$$

The initial data being supported in $\{x < 0\}$, $u = -v$, thus $u^* = -v^*$ for $x > 0$. ■

We illustrate (19) with numerical results for two different profiles $\sigma(x)$ (see figures 2 et 3).

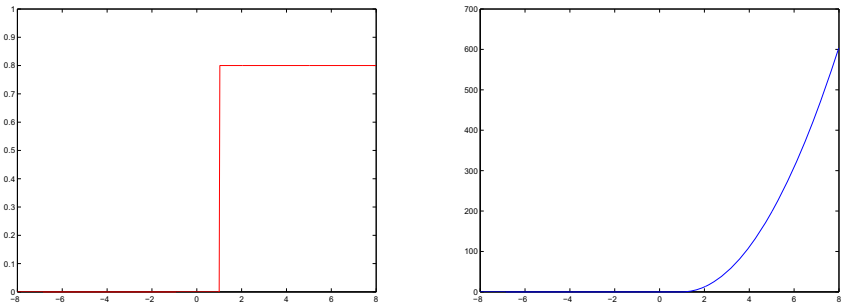


Figure 2: The profiles $\sigma(x)$

Remark 4 *If one simply follows the above proof, on could think that the system (20) (simply obtained by the change of function (19)) provides a PML. It is not the case or, more precisely, only true as long as an infinite layer in considered. Indeed, bounding the layer at $x = L$ one will generate a reflected wave propagating in the direction $x < 0$: however, if the system, (20) attenuates exponentially the waves propagating in the direction $x > 0$, it amplifies exponentially the waves propagating in the direction $x < 0$.*

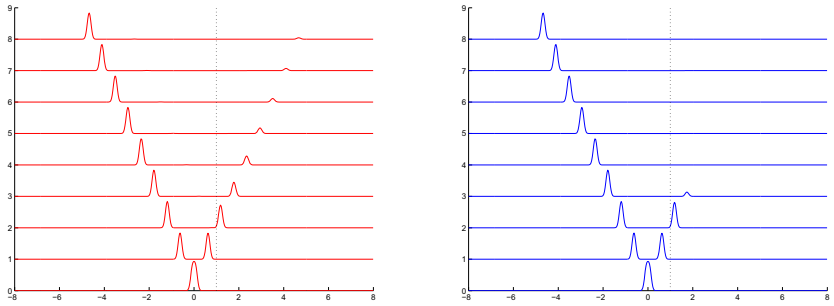


Figure 3: The solution of the Cauchy problem (infinite PML). Left: constant profile. Right: variable profile

Let us come now to the case of the finite layer. We impose, as an example, the homogeneous Neumann condition at $x = L$:

$$\frac{\partial u}{\partial x}(L,t) = 0. \tag{22}$$

Let us denote by $(\overline{u}, \overline{v})$ the solution of (14) in the domain $\{x < L\}$ (with $\rho = \mu = 1$) associated with the boundary condition (22) and initial data (u_0, v_0) with support in $\{x < 0\}$. To compute explicitly the solution, we remark the system (14) is "invariant by changing x into $-x$": if $\big(u(x,t), v(x,t)\big)$ is a solution of (14) then $\big(u(-x,t), -v(-x,t)\big)$ is also a solution. As a consequence, one has a principle of images, as for the usual wave equation. It is then easy to see that, if (u^*, v^*) is the solution of the Cauchy associated with (14) (still with $\rho = \mu = 1$) for the same initial data, one simply has

$$\left\{ \begin{array}{l} \overline{u}(x,t) = u^*(x,t) + u^*(2L - x, t) \\ \\ \overline{v}(x,t) = v^*(x,t) - v^*(2L - x, t). \end{array} \right. \tag{23}$$

The wave $u^*(2L - x, t)$ is the wave reflected by the absorbing layer. It is clear on formula (19) that this wave is exponentially attenuated in time inside the absorbing layer as it is illustrated by numerical experiments (see figure 4)

More precisely, combining (23) with (19), it is straightforward to get the following error estimate, that emphasizes the "exponential accuracy" of the PML model:

Theorem 2 *Let $(u_{\sigma,L}, v_{\sigma,L})$ the solution of the Cauchy problem with PML of width L made of (14) in the domain $\{x < L\}$ completed with the homogeneous Neumann condition at $x = L$ and initial data u_0 et v_0 with support in the half-space $x < 0$. If (u, v) denotes the solution of the corresponding Cauchy problem on the real line, one have the error estimate:*

$$\forall\, t > 0, \quad \big\|\big(u - u_{\sigma,L}\big)(\cdot,t)\big\| + \big\|\big(v - v_{\sigma,L}\big)(\cdot,t)\big\| \leq \frac{1}{2}\, e^{-\int_0^L \sigma(x)\, dx} \, \big(\|u_0\| + \|v_0\|\big), \tag{24}$$

where $\|\cdot\|$ denotes any L^p- norm $(1 \leq p \leq +\infty)$ in the half-line $x < 0$.

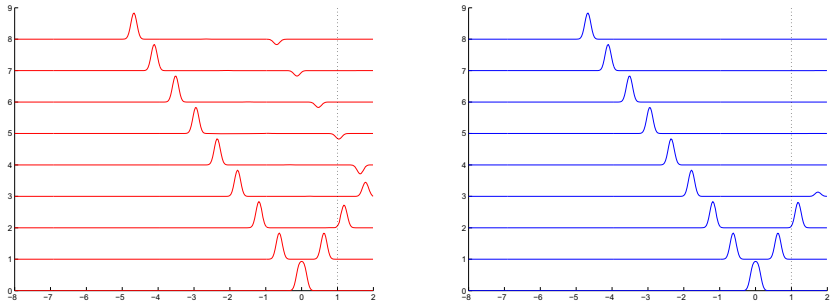


Figure 4: Finite PML layer. Left: constant profile . Right: variable profile

Remark 5 *The existence of an image principle is particularly clear on the second order equation (15) which only involves second order derivatives in x. The invariance by x into $-x$ is no longer satisfied by (20): indeed, if one eliminates u^* in (20), one gets an equation involving a first order derivative in x:*

$$\frac{\partial^2 u}{\partial t^2} - 2\sigma \frac{\partial u}{\partial x} + \sigma^2 u - c^2 \frac{\partial^2 u}{\partial x^2} = 0. \tag{25}$$

2 Perfectly Matched Layers in dimension 2

We present in this section the generalization of the PML layers introduced in the previous section to the case of problems posed in two space dimensions. The generalization to higher space dimension does not pose any additional conceptual difficulty. In the following, we shall denote x and y the two space variables and we shall be interested in constructing an absorbing layer "in the x direction", that is to say a layer that aims at reducing the computational domain to a half-space $x < L$. We shall treat the question of reducing the computational domain to a bounded region in section 2.2.

2.1 Bérenger's PML layers for a general first order hyperbolic system.

We consider a general wave propagation problem posed in $\mathbb{R}^2$ and governed by a first order hyperbolic system (most of the more relevant physical wave phenomena fall into this category)

$$\frac{\partial U}{\partial t} + \mathbf{A}_x \frac{\partial U}{\partial x} + \mathbf{A}_y \frac{\partial U}{\partial y} = 0. \tag{26}$$

In (26), the function $U(x, y, t)$ takes values in $\mathbb{R}^m, m \geq 1$, $\mathbf{A}_x$ and $\mathbf{A}_y$ are matrices $m \times m$ with real entries. To simplify our presentation, we shall assume that the propagation medium is homogeneous, i.e. that the matrices $\mathbf{A}_x$ and $\mathbf{A}_y$ are constant and that the system (26) is hyperbolic (see section 3.1).

As in the 1D case, the idea consists in concatenating the physical medium in the half-space $x < 0$ with a medium that will perfectly match and in which the waves

will be damped during their propagation. Let me introduce without any particular justification the system of equations that I will consider in the domain $x > 0$. The formal construction of these equations, that mimics the original construction of Bérenger for Maxwell's equations, can be decomposed in two steps:

(i) One splits the field U into:

$$U = U^x + U^y, \tag{27}$$

where (U^x, U^y) is solution of the system:

$$\begin{cases} \dfrac{\partial U^x}{\partial t} + \mathbf{A}_x \dfrac{\partial}{\partial x}\left(U^x + U^y\right) = 0, \\ \dfrac{\partial U^y}{\partial t} + \mathbf{A}_y \dfrac{\partial}{\partial y}\left(U^x + U^y\right) = 0. \end{cases} \tag{28}$$

In this first step, we "do not change anything" : if (U^x, U^y) is solution of (28), $U = U^x + U^y$ is solution de (26) (simply add the two equations). Reciprocally, the solution U can be decomposed as (27) where (U^x, U^y) is solution of (28) if one splits the initial data U_0 in a compatible way.

(ii) We add an absorption term $\sigma\ U^x$ with $\sigma = \sigma(x) \geq 0$ in the first equation de (28), the one which involves x derivatives, which leads to:

$$\begin{cases} \dfrac{\partial U^x}{\partial t} + \sigma\ U^x + \mathbf{A}_x \dfrac{\partial}{\partial x}\left(U^x + U^y\right) = 0, \\ \dfrac{\partial U^y}{\partial t} + \mathbf{A}_y \dfrac{\partial}{\partial y}\left(U^x + U^y\right) = 0. \end{cases} \tag{29}$$

The reader will notice that:

- One re-obtains the system (14) in the case of the 1D wave equation.
- The system (29) extends to the general system (26) the PML initially introduced by Bérenger for Maxwell's equations.
- In practice one will choose of course $\sigma = 0$ for $x < 0$ and $\sigma > 0$ for $x > 0$. Moreover, we shall not apply the decomposition (27) in the "physical" half-space $x < 0$ which means that one doubles the dimension of the unknown functions (and one can even do better for some equations, see section 2.2) only in the PML region. The two "physical" et "non physical" media will be coupled at the artificial interface $x = 0$ by the transmission condition:

$$U = U^x + U^y, \quad \text{at } x = 0. \tag{30}$$

- In the system (29), we see that the two space directions x and y are not treated in the same way. In some sense, one applies an anisotropic absorption. If we had simply added a term $\sigma\ U$ to the system (29) (which we did in 1-D), we would have created an absorbing medium (this is clear at least for symmetric systems, namely when the matrices $\mathbf{A}_x$ et $\mathbf{A}_y$ are symmetric) but this medium would not be perfectly matched (except for 1D y-independent solutions).

- Of course, to realize a PML layer in the direction y, it suffices this time to add the term $\sigma\ U^y$ to the second equation de (28), the one which involves the y derivative, the absorption coefficient σ depending this time on y.

Let us know explain how one can derive formally the equation (29) by generalizing the approach of the complex change of variable introduced in section 1.

We start from the system (26) that we rewrite in the frequency domain (formally, we apply the time Fourier transform). This gives the " generalized Helmholtz equation":

$$i\omega\ \widehat{U} + \mathbf{A}_x \frac{\partial \widehat{U}}{\partial x} + \mathbf{A}_y \frac{\partial \widehat{U}}{\partial y} = 0, \tag{31}$$

Assume that the $\widehat{U}$ can be extended to $x \in \mathbb{C}$ (this can be justified in many cases) into an analytic function. Of course, this function still satisfies (31) for x in $\mathbb{C}$. Then, let us look at the function:

$$\widetilde{U}(x) = \widehat{U}\left(x + \frac{1}{i\omega}\int_0^x \sigma(\xi)\ d\xi\right), \quad x \in \mathbb{R}$$

where the function $\sigma(x)$ has the properties that we indicated previously. Clearly, $\widetilde{U}$ coincides with $\widehat{U}$ for $x < 0$ (which guarantees perfect matching) and satisfies

$$i\omega\ \widetilde{U} + \mathbf{A}_x \frac{\partial \widetilde{U}}{\partial x}\Big(\frac{i\omega}{i\omega+\sigma}\Big) + \mathbf{A}_y \frac{\partial \widetilde{U}}{\partial y} = 0,$$

which can be rewritten, after division by $i\omega$:

$$\left|\begin{array}{rcccccl} \widetilde{U} & = & (-\dfrac{1}{i\omega+\sigma}\mathbf{A}_x\dfrac{\partial \widetilde{U}}{\partial x}) & + & (-\dfrac{1}{i\omega}\mathbf{A}_y\dfrac{\partial \widetilde{U}}{\partial y}) & , \\ & = & \widetilde{U}_x & + & \widetilde{U}_y & , \end{array}\right.$$

where by construction

$$(i\omega+\sigma)\ \widetilde{U}_x + \mathbf{A}_x\frac{\partial \widetilde{U}}{\partial x} = 0, \quad i\omega\ \widetilde{U}_y + \mathbf{A}_y\frac{\partial \widetilde{U}}{\partial y} = 0.$$

We obtain the system (29) by going back to the time variable.

At least formally, our construction allows us to understand why we realized a perfectly matched medium with the system (29). The "absorbing" nature of the new medium is much less obvious: one does not have the equivalent of theorem 1 which makes the 1D case particularly clear. One can have an intuition that this is the case (but this intuition in misleading in the general case - see remark 6), by looking at harmonic plane waves. i.e. particular solutions of (26) of the form (we assume that we are in an homogeneous medium occupying the whole space)

$$\mathbf{U}_0\ \exp i(\omega t - k_x x - k_y y), \quad (k_x, k_y, \omega) \in \mathbb{R}^3. \tag{32}$$

The hyperbolic character of the system (26) ensures the existence of such solutions, (k_x, k_y, ω) being necessarily related by the dispersion relation of (26) (of the form $P((k_x, k_y, \omega) = 0$ where P is a homogeneous polynomial in (k_x, k_y, ω)). We shall come back to this in more detail in section 3.1. For the Cauchy problem, the use of the space Fourier transform even shows that such solutions form an (infinite) *"fundamental system"* for (26).

If one adopts a physical point of view, (32) represents a plane wave with frequency ω that propagates in the direction of the wave vector $k = (k_x, k_y)$ with phase velocity:

$$c = \frac{\omega}{|k|} \cdot \frac{k}{|k|} \ . \tag{33}$$

This wave thus propagates "in the direction" $x > 0$ if $k_x/\omega > 0$ and "in the direction" $x < 0$ if $k_x/\omega < 0$. For the PML system (29), taking into account the change of variable:

$$x \to x + \frac{1}{i\omega} \int_0^x \sigma(\xi) \, d\xi,$$

the plane wave (32) "becomes":

$$U_0 \ \exp i(\omega t - k_x x - k_y y) \ e^{-\frac{k_x}{\omega} \int_0^x \sigma(\xi) \, d\xi}. \tag{34}$$

The function σ being positive, one observes then that if $k_x/\omega > 0$ (resp. $k_x/\omega < 0$), that is to say if the wave (32) propagates "in the direction" $x > 0$ (resp. $x < 0$), the wave (34) is evanescent, i. e. exponentially damped, in the direction $x > 0$ (resp. $x < 0$). In other words, the amplitude of the wave decreases in its direction of propagation. In this sense, one could say that one has realized an absorbing medium.

Remark 6 *The previous reasoning, based one the phase velocity of a single plane wave, is in fact misleading. In the 1D case, the dissipative character of the PML system, traduced in particular by the energy identity (17), went together with the absorbing nature of the medium "in the sense of plane waves", which we identify by a decay in space of such solutions inside the layer. In dimension 2, this is not always the case. In fact, when one considers a (continuous) superposition of plane waves (for instance any finite energy solution), one must reason with the group velocity (see section 3).*

Remark 7 *There is another noticeable difference with the 1D case: the absorption rate in (34) is not only proportional to σ but also k_x/ω. this means that the absorption of waves also depends on their propagation direction, which was expected from the anisotropic nature of the absorption term in (29). We shall come back to this point with the acoustic wave equation (section 2.2).*

2.2 The case of the 2D acoustic wave equation.

We start from the wave equation (which generalizes the equation (5) to the dimension 2):

$$\rho \, \frac{\partial^2 u}{\partial t^2} - \text{div} \, (\mu \nabla u) = 0, \tag{35}$$

that we rewrite in the form of a first order system

$$\left\{\begin{array}{l} \rho\,\dfrac{\partial u}{\partial t} - \dfrac{\partial v_x}{\partial x} - \dfrac{\partial v_y}{\partial y} = 0, \\ \mu^{-1}\,\dfrac{\partial v_x}{\partial t} - \dfrac{\partial u}{\partial x} = 0, \\ \mu^{-1}\,\dfrac{\partial v_y}{\partial t} - \dfrac{\partial u}{\partial y} = 0\ . \end{array}\right. \tag{36}$$

The PML system. Due to the particular structure of system (36) (two of the equations involve derivatives in only one space variable), by applying the approach of the previous paragraph, one realizes that one can avoid to split the unknowns v_x and v_y. One then obtains the following PML system :

$$\left\{\begin{array}{l} \rho\,\Big(\dfrac{\partial u^x}{\partial t} + \sigma\ u^x\Big) - \dfrac{\partial v_x}{\partial x} = 0, \\ \mu^{-1}\,\Big(\dfrac{\partial v_x}{\partial t} + \sigma\ v_x\Big) - \dfrac{\partial}{\partial x}\big(u^x + u^y\big) = 0, \\ \rho\,\dfrac{\partial u^y}{\partial t} - \dfrac{\partial v_y}{\partial y} = 0, \\ \mu^{-1}\,\dfrac{\partial v_y}{\partial t} - \dfrac{\partial}{\partial y}\big(u^x + u^y\big) = 0\ . \end{array}\right. \tag{37}$$

The reader will notice that

- Contrary to the 1D case, the dissipative character of this system, even the well-posedness of the corresponding Cauchy problem, is no longer obvious.
- When σ, ρ and μ are constant, one can eliminate the unknowns v_x and v_y and get an equation for the scalar unknown u,

  $$\Big(\frac{\partial}{\partial t} + \sigma\Big)^2\Big(\rho\frac{\partial^2 u}{\partial t^2} - \mu\frac{\partial^2 u}{\partial y^2}\Big) - \mu\ \frac{\partial^4 u}{\partial x^2 \partial t^2} = 0. \tag{38}$$

  Compared to (35), this equation becomes fourth order, which was not the case in 1D. In some sense, we have added unphysical modes.
- One can remark that the equation (38) remains invariant under the change of variable : $x \to -x$. This means that an image principle is preserved.

Absorption and reflection of plane waves. We can describe in more detail the phenomenon of absorption of plane waves already evoked in the previous section for general systems.Let us consider a homogeneous acoustic medium with velocity $c = \sqrt{\mu/\rho}$. For the equation (35) (or for system (36)),the dispersion relation reads

$$k_x^2 + k_y^2 = \frac{\omega^2}{c^2}\ . \tag{39}$$

If we denote by $\theta \in [-\pi, \pi]$ the angle between the wave vector $k = (k_x, k_y)$ and the direction $(0, 1)$, which we shall "angle of incidence" since the vector $(0, 1)$ also represents the unit normal vector to the artificial boundary $x = 0$, we thus have

$$k_x = \frac{\omega}{c} \cos\theta, \quad k_y = \frac{\omega}{c} \sin\theta. \tag{40}$$

The corresponding propagative plane (solution of (35)) writes

$$u(x,t) = \exp i \left[\frac{\omega}{c} \left(c\,t - x\cos\theta - y\sin\theta \right) \right] . \tag{41}$$

For the system (37), this wave becomes an evanescent wave

$$u(x,t) = \exp i \left[\frac{\omega}{c} \left(c\,t - x\cos\theta - y\sin\theta \right) \right] \; e^{-\frac{\cos\theta}{c} \int_0^x \sigma(\xi)\, d\xi} . \tag{42}$$

One sees that the absorption rate in the x direction (the factor of x inside the real exponential) varies proportionally $\cos\theta$. If we restrict ourselves to $|\theta| \leq \pi/2$ (waves propagating in the direction $x > 0$), this rate is maximal for $\theta = 0$, that is to say at "normal incidence", and tends to 0 when θ goes to $\pi/2$, namely when one approaches the "grazing" incidence.

If one bounds the PML layer with a Neumann boundary at $x = L$, through a simple reflection principle, one can construct, for each θ, a particular solution of the form

$$\left| \begin{array}{rcl} u(x,t) & = & \exp i \left[\frac{\omega}{c} \left(ct - x\cos\theta - y\sin\theta \right) \right] \; e^{-cos\theta \int_0^x \sigma(\xi)\, d\xi} \\ & & + \; \exp i \left[\frac{\omega}{c} \left(ct - (2L - x)\cos\theta - y\sin\theta \right) \right] \; e^{-cos\theta \int_0^{2L-x} \sigma(\xi)\, d\xi}. \end{array} \right. \tag{43}$$

Inside the physical medium $x < 0$, this solution takes the form

$$u(x,t) = \exp i \left[\frac{\omega}{c} \left(ct - x\cos\theta - y\sin\theta \right) \right] + R_\sigma(\theta) \; \exp i \left[\frac{\omega}{c} \left(ct + x\cos\theta - y\sin\theta \right) \right] , \tag{44}$$

where we have set

$$R_\sigma(\theta) = \exp \left[-\frac{2\cos\theta}{c} \int_0^L \sigma(\xi)\, d\xi \right] \; \exp \left[-2i \frac{\omega L}{c} \right] . \tag{45}$$

This can be interpreted in the following way : the incident wave (41) gives rise on the boundary $x = 0$ to a reflected wave which obeys the standard Snell-Descartes' law for the propagation direction, butis affected with a phase shift $2\omega L/c$ (corresponding to a time delay due to the back and forth propagation inside the layer) and has an amplitude given by the reflection coefficient:

$$|R_\sigma(\theta)| = \exp \left[-2 \frac{\cos\theta}{c} \int_0^L \sigma(\xi)\, d\xi \right] . \tag{46}$$

One sees that this reflection coefficient diminishes exponentially with respect to the absorption coefficient σ, the width L of the layer, and the cosine of the angle of incidence θ.

Remark 8 *The expression (45) can be compared with the one obtained with Engquist-Majda absorbing boundary conditions of order N (an integer greater or equal than 1),*

$$|R_N(\theta)| = \left(\frac{1-\cos\theta}{1+\cos\theta}\right)^N. \tag{47}$$

Why choosing a σ that varies with x ? In practice, after discretization of the problem for numerical computation, if one considers a fixed space step in the direction x inside the layer, the additional computational cost due to the PML layer is directly proportional to the width L : one thus have interest to reduce L as much as possible. If one refers to the expression (45) of the reflection coefficient, this gives the feeling that it suffices to choose $\sigma > 0$ constant and arbitrarily large to be able to take L arbitrarily small ! In practice, this is not true because one has to work with a numerical approximation of the wave equation. As a consequence, if the layer is effectively perfectly matched for the continuous problem, this is no longer the case for the discrete problem. This has been analyzed [11] when one considers the standard leap-frog scheme with centered second order finite differences. It is not my objective to come back here in detail the results of this analysis and I will simply describe qualitatively the main conclusions :

- As the layer is no longer perfectly matched at the discrete level, for the numerical scheme an incident plane wave will give rise to two reflected waves (see figure 5):
  - a wave reflected at $x = L$, which I will call the PML reflected wave,
  - a wave reflected at the interface $x = 0$, which I will call the numerical reflected wave.

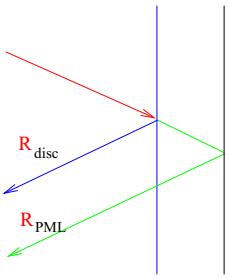


Figure 5: Discrete reflections : the case of a constant profile.

- The PML reflected wave, of amplitude R_{PML}, is of the same nature que than the continuous reflected wave. It results from the back and forth travel of a wave that propagates inside the layer and is exponentially attenuated during its propagation. More precisely, Δx denoting the space step, as the numerical scheme is second order accurate

$$R_{PML} = e^{-2\frac{\sigma L}{c}\cos\theta}\big(\,1 + O(\Delta x^2)\,\big).$$

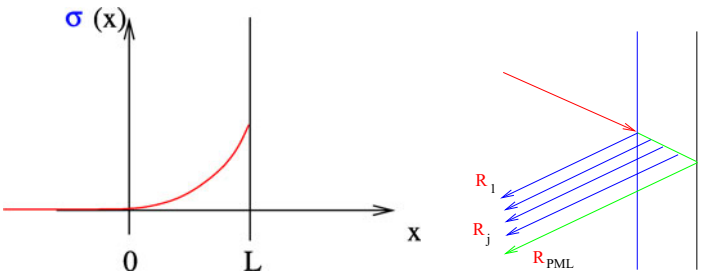
- The amplitude R_{disc} of the numerical reflected wave satisfies

$$R_{disc} \sim Cte \times \sigma^2 \Delta x^2, \quad (\Delta x \to 0).$$

  In some sense, one goes back to the dilemma encountered with classical absorbing layers:

  - as expected, the amplitude of the numerical reflected tends to 0 with Δx (proportionnally to Δx^2 because the scheme is second order accurate),
  - this amplitude increases quadratically with σ: in other words, the greater is σ, the less adapted is the absorbing layer at the discrete level.

There is a rather natural idea to overcome this difficulty. This consists is choosing a (sufficiently smooth) profile $\sigma(x)$ that progressively increases with x starting from $\sigma(0) = 0$. It is easy to get some intuition about what will happen at the numerical level, which we try to illustrate of figure 6. In practice the discretization processs is equivalent to create successive thin (i. e. of width Δx) homogeneous layers with absorption σ_j inside the jth layer. As a consequence, to the "continuous" usual PML reflected wave, one adds the superposition of small (and purely numerical) reflected waves at each interface, whose amplitudes will be proportional to the successive jumps $\sigma_{j+1} - \sigma_j$. Moreover, the amplitude of these numerical reflected waves will be smaller and smaller when j increases because of the exponential decay phenomenon. Because of this observation, it is natural and a priori not penalizing to chooze a profile σ such that the jumps $\sigma_{j+1} - \sigma_j$ are increasing with j, which corresponds to take $\sigma(a)$ as an increasing and convex function. As a matter of fact, many authors suggest to chooze a quadratic profile.



$$R_1 \sim (\sigma_1 \Delta x)^2, R_j \sim (\sigma_{j+1} - \sigma_j)^2 \Delta x^2 \rho_j, \quad \rho_j = \exp(-\frac{2}{c} \int_0^{x_j} \sigma(\xi)\, d\xi).$$

Figure 6: Discrete reflections : the case of a vaiable profile

Let us try to illustrate this with three numerical experiments. In the figure 7 we compare three computations. The first two computations are realized with the same constant σ, the second corresponding to a twice finer mesh than the first one. The reflected waves that one "sees" on the pictures 'correspond to the numerical reflected wave (observe for instance that there is no phase shift at the interface): the amplitude of this wave diminishes when one refines the mesh. The third computation is done with a quadratic profile $\sigma_{quad}(x)$ such that

$$\int_0^L \sigma_{quad}(x)\, dx = \sigma\, L$$

so that the continuous coefficient reflection is the same as for the constant profile (see (46)) : the numerical reflected wave has disappeared, even though the coarse mesh has been used. These observations are confirmed on the figure 8 where one looks at the evolution of the solution at a given point (inside the physical domain) as a function of time. The red and blue curves correspond to the constant profile, respectively for the coarse and the fine mesh. On the other hand, one can check that the green curve fits perfectly with the exact solution. Once again, one clearly sees the numerical reflected wave that decays when one refines the mesh.

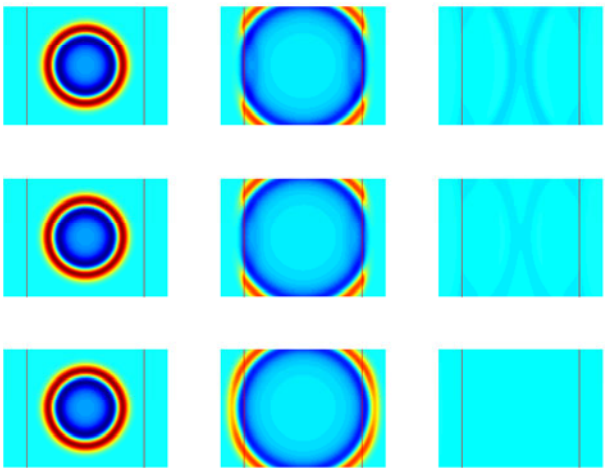


Figure 7: Top : constant profile and "coarse" mesh. Middle : constant profile and "coarse" mesh. Bottom : quadratic profile and "coarse" mesh. The level lines of the solution are represented at three consecutive instants (left to right)

Remark 9 *One can of course, for a given length L (chosen proportionally to Δx to reason at constant computational cost) and a given numerical scheme, try to optimize*

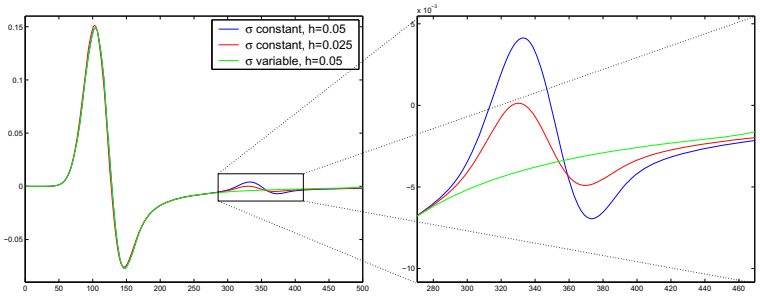


Figure 8: The solution at a given point as a function of time. Blue: constant profile and "coarse" mesh. Red: constant profile and "fine" mesh. Green: quadratic profile and "coarse" mesh.

the profile $\sigma(x)$ in order to minimize the reflections (typically for a given range of frequencies). This has been done for instance in [11]. In practice, implementing such a process is quite constraining and one will prefer so impose a profile $\sigma(x)$ given a priori (possibly depending on Δx and on the numerical scheme). We refer the reader to the abundant literature for more concrete information.

How to reduce the computational domain to a rectangle ? To reduce effective calculations to a bounded domain, it may be sufficient in some cases (in a waveguide for instance) to use only straight boundaries (the other boundaries being of physical nature). However in most cases, exterior problems for instance, this is no longer sufficient, "all the boundaries" of the computational domain needing to be absorbing. The method of PML's that we have presented easily allows us to reduce "easily" the reduction to a rectangle. Let us assume that the physical domain of interest is the rectangle $[-a, a] \times [-b, b]$. We are going to introduce an absorbing layer of width L and pose a problem in the rectangle $[-a-L, a+L]\times[-b-L, b+L]$ with the following system of equations:

$$\left\{\begin{array}{l} \rho\left(\dfrac{\partial u^x}{\partial t} + \sigma_x(x)\ u^x\right) - \dfrac{\partial v_x}{\partial x} = 0, \\ \mu^{-1}\left(\dfrac{\partial v_x}{\partial t} + \sigma_x(x)\ v_x\right) - \dfrac{\partial}{\partial x}\left(u^x + u^y\right) = 0, \\ \rho\left(\dfrac{\partial u^y}{\partial t} + \sigma_y(y)\ u^y\right) - \dfrac{\partial v_y}{\partial y} = 0, \\ \mu^{-1}\left(\dfrac{\partial v_y}{\partial t} + \sigma_y(y)\ v^y\right) - \dfrac{\partial}{\partial y}\left(u^x + u^y\right) = 0\ . \end{array}\right. \tag{48}$$

where the positive function σ_x (resp. σ_y) only depends on x (resp. y), has support in $\{|x-a| > L\}$ (resp. $\{|y-b| > L\}$) and ensures the absorption in the x (resp. y) direction. The equations (48) can easily be obtained by repeating the process described in section 2.1, first in the x direction, next in the y direction. This process

provides an automatic and particularly simple (in comparison with what one needs to do with absorbing boundary conditions) treatment of the corners of the computational domain. We illustrate in figure 9 the type of result that can be obtained : in the corresponding experiment, the acoustic wave is emitted by a point source located at the center of the computational domain, which is a square in this case (the physical domain corresponding to the "interior" square).

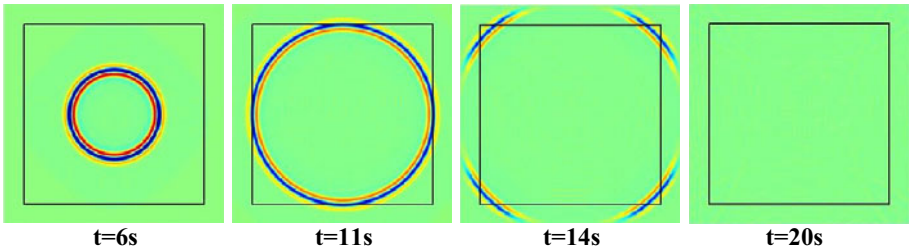


Figure 9: Propagation of a 2-D acoustic wave with PMLs.

Remark 10 *PML's are thus well adapted to rectangular boundaries which is particularly useful in one uses finite difference methods for the discretization. F. Collino et P. Monk have treated a circular boundary in [12]. The case of "arbitrary" curved boundaries is considered in [30].*

3 Mathematical questions

Up to now, we did not attack the purely theoretical questions linked to PMLs. As soon as these techniques have appeared, many researchers have been interested in their mathematical analysis. A first article by Abarbanel and Gottlieb [1] (claiming the the PML model in electromagnetism was "ill-posed", I will come back on this) provoked many interrogations (in particular because their statement seemed to be in contradiction with numerical observations). Since then, much progress has been made in understanding the theoretical properties of PMLs. This chapter will treat some aspects of this topic.

The first natural question is : does the PML model give rise to a well-posed evolution problem ? The same question was posed of course for absorbing boundary conditions and the answer was far from obvious (see remark 11). In what follows, we shall be interested in the Cauchy problem associated the system (29) (for simplicity, we restrict ourselves to the PML model "in the x direction"). At least, we would like to investigate the well-posedness issue in the very simple situation where σ is constant and non zero in all $\mathbb{R}^2$. This question will be addressed in most of this section. More precisely, two distinct aspects (I insist on this because the distinction is not always done properly in the literature, where one there is sometimes confusion between the words "ill-posed" and "unstable") deserve to be treated the one after the other :

- the well-posedness : obtain an existence an uniqueness result for the solution, with continuous dependence with respect to the data of the problem,
- the stability, which consists essentially in studying the long time behaviour of the solution.

As the system (29) appears as a perturbation of order zero of a first order homogeneous system, it is useful to recall in a first step some classical results from the theory of this type of system.

Remark 11 *In the case of the absorbing boundary conditions, one has to analyze what one usually calls a "mixed" hyperbolic problem, namely an initial boundary problem for an hyperbolic system, which is a more difficult problem than the pure Cauchy problem that we investigate here. The most "standard" approach consists is applying the Kreiss theory ()[26], [21]), or "Normal Mode Analysis", which appears as a generalization (rather complex in its most technical details) of the Fourier analysis (that we shall present in the next section) for the Cauchy problem. The application of this theory shows that the problem under consideration is ill-posed when the absorbing boundary condition is not constructed properly [16] and leads to rather simple algebraic criteria, in the case of the wave equation for instance [31], that guarantee well-posedness. An alternative is the use of energy methods ([24, 25]), which are quite tricky in this context.*

3.1 A recap of some classical results.

We consider a Cauchy problem in $\mathbb{R}^2$ of the form:

$$\left\{ \begin{array}{l} \dfrac{\partial V}{\partial t} + A_x \dfrac{\partial V}{\partial x} + A_y \dfrac{\partial V}{\partial y} + BV = 0, \quad (x,y) \in \mathbb{R}^2, \quad t > 0, \\ V(x,y,0) = V_0(x,y), \quad (x,y) \in \mathbb{R}^2, \\ \\ V \in \mathbb{R}^d, (A_x, A_y, B) \in \mathcal{L}(\mathbb{R}^d)^3. \end{array} \right. \tag{49}$$

Note that (29) enters this framework with $d = 2m$ and:

$$V = \begin{pmatrix} U^x \\ U^y \end{pmatrix}, \quad A_x = \begin{pmatrix} \mathbf{A}_x & \mathbf{A}_x \\ 0 & 0 \end{pmatrix}, \quad A_y = \begin{pmatrix} 0 & 0 \\ \mathbf{A}_y & \mathbf{A}_y \end{pmatrix}, \quad B = \begin{pmatrix} \sigma I & 0 \\ 0 & 0 \end{pmatrix}.$$

The natural framework for this type of problem is the L^2 framework and related Sobolev spaces in $\mathbb{R}^2$, which shall be denoted $H^s, s \in \mathbb{R}$. Most of what follows is quite classical and we refer the reader to [27] for more details.

In what follows, I will make the distinction between the *problem* (49) (the Cauchy problem) and the *system* (49) (namely the first line of (49) only).

Definition 1 (Well-posedness) *We shall say that the problem (49) is well posed if and only if there exists $s \geq 0$ such that for any V_0 in H^s, (49) admits a unique solution:*

$$V \in C^0(\mathbb{R}^+, L^2) \tag{50}$$

and satisfies an estimate of the form (where $C(t)$ denotes a positive function of t, independent of V_0, whose behavior for large t is not prescribed) :

$$\forall\, t > 0, \quad \|V(t)\|_{L^2} \leq C(t)\, \|V_0\|_{H^s}\ . \tag{51}$$

More precisely:

- *One says that (49) is strongly well-posed if $s = 0$. In this case, there is no loss of (space) regularity with respect to the initial condition.*
- *One says that (49) is weakly well-posed if one can obtain (51) only for some $s > 0$.*

Finally, we shall say that the problem is ill-posed if it is not well posed.

The Fourier analysis. In the case of constant coefficients, the simplest method to study (49) consists in using the Fourier transform in space

$$V(x, y, t) \longrightarrow \widehat{V}(k_x, k_y, t),$$

which immediately provides the expression ($k := (k_x, k_y)$):

$$\widehat{V}(k, t) = \exp\big[\, -i\, \big(A(k) - iB\big)\, t\,\big]\ \ \widehat{V}_0(k),$$

where the "symbol" matrix $A(k)$ is given by:

$$A(k) = k_x A_x + k_y A_y.$$

Obtaining estimates of the type (51) relies of Plancherel-Parceval's theorem and estimates of the exponentials of matrices:

$$\exp\big[\, -i\, \big(A(k) - iB\big)\, t\,\big] \tag{52}$$

of which we need in particular to control the behavior for large values of $|k|$ (which will determine the value of s in (51)). One can show that the study of (52) amounts to the study of particular solutions of (49) which are harmonic plane waves or *normal modes*:

$$V(x, y, t) = V\ \exp i(k_x x + k_y y)\ e^{i\omega t}, \quad k = (k_x, k_y) \in \mathbb{R}^2, \quad \omega(k) \in \mathbb{C}, \quad V \in \mathbb{C}^m. \tag{53}$$

Substituting (53) into (49) gives :

$$\big(A(k) - iB + \omega\big)\, V = 0, \tag{54}$$

In other words, to get a non trivial solution, $-\omega$ must be an eigenvalue of $A(k) - iB$, V being the corresponding eigenvector. This leads to the *dispersion relation*

$$\det\,(A(k) - iB + \omega I) = 0, \tag{55}$$

that is considered here as an equation in ω, k being a parameter. Denoting by

$$\{\ \omega_j(k),\ 1 \leq j \leq d\ \}$$

the set of branches of solutions de (55) (namely the eigenvalues of $iB - A(k)$, which gives d branches of solution - if one takes into account the possible multiplicity of these eigenvalues), it is easy to see that (49) is (weakly or strongly) well posed (in the sense of definition 1) if and only if, for all j

$$\mathcal{I}m\ \omega_j(k) \text{ remains } \textit{bounded from below}, \quad \forall\ k \in \mathbb{R}^2. \tag{56}$$

Geometrically, this means that all the curves in the complex plane (they form a family of curves parameterized by j and the unit vector $K \in \mathbb{R}^2,\ |K| = 1$, which represents a direction) :

$$|k| \mapsto \omega_j\big(|k| \cdot K\big), \quad K = k/|k|, \tag{57}$$

are located *above* some complex half-space. As the functions $\omega_j(k)$ are continuous (from classical results from perturbation analysis in spectral theory), in order to ensure (56), it is sufficient to study what happens when $|k|$ tends to $+\infty$.

The case of the unperturbed system ($B = 0$). The results for the problem (49) can be obtain by perturbation techniques with respect to the case $B = 0$, which corresponds to the first order homogeneous part of the system (49), that we shall call the unperturbed system (this is rather intuitive, note for instance that, for large $|k|$, $A(k) - iB$ behaves as $A(k)$). Let us recall below the standard terminology for such systems.

Definition 2 (Hyperbolicity) *The unperturbed system ($B = 0$), denoted (49)$_0$, is said to be hyperbolic if and only if :*

- $\forall\ k \in \mathbb{R}^2, \quad$ *the eigenvalues of $A(k)$ are real.*

It is said strongly hyperbolic if in addition (in the opposite case, it is said weakly hyperbolic):

- $\forall\ k \in \mathbb{R}^2, \quad A(k)$ *is diagonalizable.*

Remark 12 *An important class of strongly hyperbolic systems is provided by symmetric hyperbolic systems that correspond to the case where*

$$A_x \textit{ and } A_y \textit{ are real symmetric matrices.}$$

This class of systems includes most of the classical wave propagation models from the physics.

When $B = 0$, (54) simply means that $-\omega$ must be an eigenvalue of $A(k)$. Thus, the functions $\omega(k)$ are homogeneous functions of degree 1 with respect to k and the curves (57) are simply half-lines. Moreover, $A(k)$ being real, its non real eigenvalues can be regrouped by conjugated pairs. The half-lines corresponding to the real eigenvalues are

on the real axis while the other ones can be regrouped by pairs in which each half-line is not included in the real axis and deduced from the other by symmetry with respect to the real axis, which means that for one of them the imaginary part goes to $-\infty$, making (56) impossible. As a consequence, we get a well-posed problem if and only if all the half-lines are on the real axis, which is equivalent to the hyperbolicity of $(49)_0$. One has in fact the following more precise result:

Theorem 3 *When $B = 0$, the Cauchy problem (49) is well-posed if and only if the system $(49)_0$ is hyperbolic. Moreover*

$(49)_0$ is strongly hyperbolic $\Rightarrow$ (49) is strongly well-posed and

$$(E1) \quad \forall\, t > 0, \quad \|U(t)\|_{L^2} \leq C\, \|U_0\|_{L^2}$$

$(49)_0$ is weakly hyperbolic $\Rightarrow$ (49) is weakly well-posed and

$$(E2) \quad \forall\, t > 0, \quad \|U(t)\|_{L^2} \leq C\,(1+t)^s\, \|U_0\|_{H^s}, \quad s \in \mathbb{N}, \quad s \geq 1$$

where $s + 1 = \sup_k$ *max* $\{$*Sizes of Jordan blocks of* $A(k)\}$.

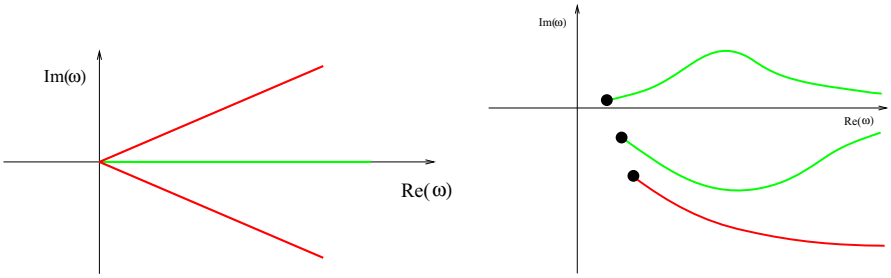


Figure 10: The branches $\omega(k)$. Left, the hyperbolic unperturbed system. Right, the perturbed case (in red, a "strongly unstable" mode)

The case of the perturbed system. Obviously, we assume (this is a necessary condition for well-posedness) that the unperturbed system is hyperbolic. Intuitively, one understand that the half-lines corresponding to the unperturbed case are asymptotic branches, when $|k| \to +\infty$, for the curves $|k| \mapsto \omega\big(|k| \cdot K\big)$. This is the basic idea for proving the following theorem.

Theorem 4 *If the unperturbed system $(49)_0$ is strongly hyperbolic, the (49) is strongly well-posed and there exists a positive constant $M \geq 0$ such that:*

$$\forall\, t > 0, \quad \|V(t)\|_{L^2} \leq C\, e^{Mt}\, \|V_0\|_{L^2}\ . \tag{58}$$

If the unperturbed system $(49)_0$ is weakly hyperbolic, then, for certain matrices B, the problem (49) is ill posed.

If the unperturbed system $(49)_0$ is only *weakly* well posed, the perturbed problem (49) is not necessarily ill-posed : this will depend on the perturbation matrix B (see the simple 1D example treated at the end of this section). The case where (49) is ill-posed corresponds to the existence of branches of solutions of (55) satisfying, at least for certain directions K,

$$\mathcal{I}m\,\omega\big(|k|\,K\big) \sim -\,\gamma(K)\,|k|^{\frac{1}{s+1}}, \quad \gamma(K) > 0, \tag{59}$$

where the integer s is defined as in Theorem 3. These are what I will call "strongly unstable" modes. Geometrically, they correspond to the existence of curves (57) admitting a "parabolic branch" which goes to infinity in the complex half-space $\mathcal{I}m\,\omega < 0$ (see figure 10).

The notion of well-posed problem does not exclude solutions that are exponentially increasing in time. This notion is thus not sufficient to represent the behavior that one expects for an absorbing medium : a PML layer generating exponentially increasing solutions would be useless ! That is why it is useful to introduce the notion of stability which expresses that the solution of the perturbed problem can be controlled for large time as the solution of the corresponding unperturbed hyperbolic problem.

Definition 3 (Stability) *We assume here that the Cauchy problem (49) is well posed. One says then that (49) is strongly stable if the solution satisfies the estimate (E1) and weakly stable if it satisfies the estimate (E2) (see theorem 3). Otherwise, we say that (49) is unstable.*

Using Fourier analysis, it is easy to see that, for all j:

$$(49) \text{ is (weakly or strongly) stable } \iff \mathcal{I}m\,\omega_j(k) \geq 0, \quad \forall\, k \in \mathbb{R}^2. \tag{60}$$

Let us conclude this section of recaps by an example in dimension 1 (x is the only space variable) with $d = 3$, which illustrates what we said in a simple way:

$$\frac{\partial}{\partial t}\begin{pmatrix} u \\ v \\ w \end{pmatrix} + A\,\frac{\partial}{\partial x}\begin{pmatrix} u \\ v \\ w \end{pmatrix} + B\begin{pmatrix} u \\ v \\ w \end{pmatrix} = 0. \tag{61}$$

where the matrix A is given by :

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & -1 & 0 \end{pmatrix},$$

One easily checks that A, whose eigenvalues are 0 (simple) et 1 (double), is not diagonalizable. Let us consider successively ($a \in \mathbb{R}$):

$$B = B_1 = \begin{pmatrix} 0 & -1 & 0 \\ a & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad B = B_2 = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & a \\ 0 & 0 & 0 \end{pmatrix}.$$

With $B = B_1$, (61) is ill-posed. To be convinced, it suffices for instance to remark that, if one eliminates v et w, one gets the following scalar equation for u

$$\frac{\partial}{\partial t}\left(\frac{\partial^2 u}{\partial t^2} + 2\frac{\partial^2 u}{\partial x \partial t} + \frac{\partial^2 u}{\partial x^2} + a\frac{\partial u}{\partial x}\right) = 0. \tag{62}$$

The solutions of the dispersion equation are

$$\omega = 0, \quad \omega = -k \pm |ak|^{\frac{1}{2}}\ \frac{1+i}{\sqrt{2}}. \tag{63}$$

The imaginary part of one of them goes to $-\infty$ (in $|k|^{\frac{1}{2}}$) when $|k|$ tends to $+\infty$.

With $B = B_1$, (61) is well-posed. Indeed, the scalar equation for u writes:

$$\frac{\partial}{\partial t}\left(\frac{\partial^2 u}{\partial t^2} + 2\frac{\partial^2 u}{\partial x \partial t} + \frac{\partial^2 u}{\partial x^2} + au\right) = 0. \tag{64}$$

The solutions of the dispersion equation are thus

$$\omega = 0, \quad \omega = -k \pm \sqrt{a}. \tag{65}$$

Their imaginary part is constant (independent of k) thus bounded when k varies. Moreover, one also sees that the system is stable if $a \geq 0$ and unstable if $a < 0$ (one of the ω's has a negative imaginary part).

Remark 13 *The well-posedness and stability of (64) can also be studied with energy techniques.*

3.2 On the well-posedness of PML models

As I already said, the article by Abarbanel and Gottlieb [1] gave a "warning statement" by remarking that Bérenger's PML model for Maxwell's equations was a perturbation of a weakly hyperbolic system (this observation is in fact valid for "all equations" , see [6]). At the same time, many engineers and computer scientists were able to use this technique without meeting any particular problem with their computations. The reason is that the 0-order perturbation proposed by the PML model is not one of those which make the problem ill-posed.

The case of the wave equation. Let us consider the system (37) with constant σ and $\rho = \mu = 1$. As one easily sees on equation (38), the dispersion relation of (37) is simply

$$\det\left(A(k) - iB + \omega I\right) = (\omega^2 - k_y^2)(i\omega + \sigma)^2 + \omega^2 k_x^2 = 0, \tag{66}$$

and the one of the unperturbed system, i. e. for $\sigma = 0$ (the equations for the eigenvalues of $A(k)$), reads

$$\omega^2\left(\omega^2 - k_x^2 - k_y^2\right) = 0. \tag{67}$$

One sees in particular that $\omega = 0$ is a double eigenvalue of $A(k)$ and one easily checks that $A(k)$ is not diagonalizable. To be sure that the PML system (37) is nevertheless well-posed, it suffices to make an asymptotic expansion for large values of $|k|$ of the four branches of solutions $\omega(k)$ of (66). One easily obtains

$$\begin{cases} \omega(k) = \pm\,|k| + i\sigma\; k_x^2/|k|^2 + O\big(|k|^{-1}\big), \\ \\ \omega(k) = \sigma\; \dfrac{\pm k_x k_y + i k_y^2}{|k|^2} + O\big(|k|^{-1}\big), \end{cases} \tag{68}$$

which shows that the imaginary part of each $\omega(k)$ remains bounded (in particular bounded from below) when $|k|$ goes to infinity, whatever is the sign of σ, implying the (weak) well-posedness of the Cauchy problem. Moreover, the imaginary part has the same sign than σ which means that, as expected, the stability of the Cauchy problem depends on the sign of σ. In fact, it is not too difficult to prove that

$$\sigma \geq 0 \quad \Longrightarrow \quad 0 \leq \mathcal{I}m\; \omega(k) \leq \sigma, \quad \forall\; k \in \mathbb{R}^2, \tag{69}$$

and establish, with some more technical work, the following theorem

Theorem 5 *[10] The Cauchy problem associated with (37) with constant coefficients is weakly well-posed. More precisely, for any given initial data* $V_0 = (u_0^x, u_0^y, v_{x,0}, v_{y,0})$ *in* H^1*, there exists a unique solution:*

$$(u = u^x + u^y, v_x, v_y) \in C^0(\mathbb{R}^+, H^1), \quad e = u^x - u^y \in C^0(\mathbb{R}^+, L^2),$$

and, when $\sigma > 0$*, one has the following estimates (that show the (weak) stability of the problem)*

$$\begin{cases} \|u(.,t)\|_{L^2} + \|v_x(.,t)\|_{L^2} + \|v_x(.,t)\|_{L^2} \leq\; C\; \|V_0\|_{L^2}, \\ \\ \|e(.,t)\|_{L^2} \leq\; C\; \|V_0\|_{H^1}\;. \end{cases} \tag{70}$$

Remark 14 *Let us notice that the loss of one order of regularity (for the solution with respect to the initial data) and the corresponding linear increase of the solution only affects the "unphysical" field* e*. From this point of view, the problem (37) is "almost" strongly well-posed.*

When σ *is negative, one obtains analogous estimates with* $C \exp |\sigma| t$ *instead of* C*. As a matter of fact, in this case, the medium is no longer absorbing and the solutions are exponentially increasing (in other words the estimate is sharp).*

The proof of theorem 5 given in [10] is a direct application of Fourier analysis. Other authors showed the weak well-posedness of the Cauchy problem by "rewriting" as a zero order perturbation of a strongly hyperpolic system, via a change of unknown functions (see remark 1). This is for instance the case of the Zhao-Cangellaris formulation on which I will come back later. However, this type of approach, alone, does not permit to put in evidence the role of the sign of σ in the analysis (see remark 14). We are now going to see that the stability of the Cauchy problem, emphasized by the estimates (70), is also the consequence of an energy dissipation result.

Remark 1 *Once the problem is written as a perturbation of a strongly hyperbolic problem, one can apply theorem 4 and deduce the existence of estimates of the type (58). The "new PML system" (the articles often evoke new PMLs, with respect to Bérenger ones: they are in my opinion the "same" PMLs formulated differently) appears as the coupling between a first order strongly hyperbolic system, whose dimension (I mean here the number of scalar equations) is the same that the one of the original physical system, and several additional ordinary differential equations corresponding to the introduction of new unknowns. The new system is a priori better adapted, in terms of robustness, to numerical approximation (which is another important motivation). Such an approach has been derived first for the Maxwell's system in [32], [28] et systematized in [29]. From one formulation to the other, some mathematical properties of the system may change but this will affect only the behavior of the "non physical" unknowns (see for instance [3] or [7] for the questions related to long time behavior).*

For symmetric hyperbolic systems, of which (36) is an example, the alternative technique to analyze the evolution problem in the energy method. For (36), one has the well known energy conservation result:

$$\frac{d}{dt}\Big\{\frac{1}{2}\int \left(\rho\,|u|^2+\mu^{-1}|v|^2\right)\,dx\Big\}=0. \tag{71}$$

It is thus natural to wonder whether analogous energy estimates are available for the perturbed system (37).

Remark 2 *The same question is investigated [24], [25] for absorbing boundary conditions. One shows that the well-posedness of the mixed hyperbolic problem "wave equation + absorbing boundary conditions" always corresponds to a dissipation result for a certain energy.*

We are going to see that this is effectively the case by using the Zhao-Cangellaris formulation of (37). We introduce the change of unknowns:

$$(u^x,u^y,v_x,v_y)\to(u,v_x,v_y,v_y^*)$$

where u et v_y^* are characterized by:

$$u=u^x+u^y,\quad \frac{\partial v_y^*}{\partial t}=\frac{\partial v_y}{\partial t}+\sigma v_y\ . \tag{72}$$

To obtain the new system in (u,v_x,v_y,v_y^*), one first keeps the equations 2 and 4 of (37) (which can directly be written in terms of the new unknowns). Next, from the the equations 1 and 3 of (37), we deduce

$$\left\{\begin{array}{l}\displaystyle\frac{\partial}{\partial t}\left[\rho\,(\frac{\partial u^x}{\partial t}+\sigma u^x)-\frac{\partial v_x}{\partial x}\right]=0,\\[2ex]\displaystyle\Big(\frac{\partial}{\partial t}+\sigma\Big)\left[\rho\,\frac{\partial u^y}{\partial t}-\frac{\partial v_y}{\partial y}\right]=0.\end{array}\right.$$

Adding these two equations, we get

$$\frac{\partial}{\partial t}\left(\rho\left(\frac{\partial}{\partial t}+\sigma\right)u-\frac{\partial v_x}{\partial x}\right)-\frac{\partial}{\partial y}\Big(\frac{\partial}{\partial t}+\sigma\Big)\,v_y=0.$$

Using the equation:

$$\frac{\partial v_y^*}{\partial t}=\frac{\partial v_y}{\partial t}+\sigma v_y, \tag{73}$$

one eliminates v_y for v_y^* and, after time integration (we omit the details, initial conditions have to be taken into account, see remark 15), we get:

$$\rho\,\frac{\partial u}{\partial t}+\sigma u+\frac{\partial v_x}{\partial x}+\frac{\partial v_y^*}{\partial y}=0. \tag{74}$$

One finally obtains the new system by joining (74) and (73) to the equations 2 and 4 of (37):

$$\begin{cases} \rho\dfrac{\partial u}{\partial t}+\sigma u+\dfrac{\partial v_x}{\partial x}+\dfrac{\partial v_y^*}{\partial y}=0,\\[2mm] \mu^{-1}\,\Big(\dfrac{\partial v_x}{\partial t}+\sigma v_x\Big)-\dfrac{\partial u}{\partial x}=0,\\[2mm] \mu^{-1}\,\dfrac{\partial v_y}{\partial t}-\dfrac{\partial u}{\partial y}=0,\\[2mm] \dfrac{\partial v_y^*}{\partial t}=\ \dfrac{\partial v_y}{\partial t}+\sigma\ v_y. \end{cases} \tag{75}$$

One sees that one has coupled a system of three partial differential equations (that looks like (36)) with one ordinary differential equation.

Remark 15 *The equivalence between the Cauchy problems associated to (37) and (75) requires compatibility conditions between the initial data (see [10]).*

Theorem 6 *The solution of the Cauchy problem associated with (75) satisfies the identity:*

$$\begin{aligned} &\frac{1}{2}\frac{d}{dt}\left\{\int\rho\,\Big|\frac{\partial u}{\partial t}+\sigma u\Big|^2\,dx+\int\mu^{-1}\,\left(\Big|\frac{\partial v_x}{\partial t}\Big|^2+\Big|\frac{\partial v_y^*}{\partial t}\Big|^2\right)\,dx\right\}\\ &\qquad+\frac{d}{dt}\left\{\int\sigma^2\mu^{-1}|v_x|^2\,dx\right\}+2\int\sigma\,\mu^{-1}\,\Big|\frac{\partial v_x}{\partial t}\Big|^2\,dx=0. \end{aligned} \tag{76}$$

Proof. It is quite simple. Let us apply the operator $\partial/\partial t$ to the first equation of (75) and the operator $\partial/\partial t+\sigma$ to the next two equations. Then, one uses the fourth equation or (75) to eliminate (as already done above) v_y for v_y^*. We get :

$$\begin{cases} \rho\dfrac{\partial}{\partial t}\Big(\dfrac{\partial u}{\partial t}+\sigma u\Big)+\dfrac{\partial}{\partial x}\Big(\dfrac{\partial v_x}{\partial t}\Big)+\dfrac{\partial}{\partial y}\Big(\dfrac{\partial v_y^*}{\partial t}\Big)=0,\\[2mm] \mu^{-1}\,\Big(\dfrac{\partial^2 v_x}{\partial t^2}+2\sigma\dfrac{\partial v_x}{\partial t}+\sigma^2 v_x\Big)-\dfrac{\partial}{\partial x}\Big(\dfrac{\partial u}{\partial t}+\sigma u\Big)=0,\\[2mm] \mu^{-1}\,\dfrac{\partial^2 v_y^*}{\partial t^2}-\dfrac{\partial}{\partial y}\Big(\dfrac{\partial u}{\partial t}+\sigma u\Big)=0. \end{cases} \tag{77}$$

Next, we multiply the first equation of (77) by $\partial u/\partial t + \sigma u$, the seconde one by $\partial v_x/\partial t$ and the third one by $\partial v_y^*/\partial t$. We add the three resulting equalities and integrate in space over $\mathbb{R}^2$. After integration by parts, space derivatives disappear and one obtains (76). ■

The identity (76) does emphasize the dissipative character of the PML system when $\sigma > 0$ (and its anti-dissipative character when $\sigma < 0$!). Of course, it also leads to stability estimates.

Remark 16 *The result of theorem is no longer valid when σ varies in space (however ρ and μ can vary !). Indeed, if the system (75) is still valid, the system (77) is not because I have implicitly used the commutation between the operators:*

$$\frac{\partial}{\partial t} + \sigma \; et \; \frac{\partial}{\partial x}.$$

It is still possible to derive a priori estimates with a variable σ by energy techniques ([10]) but these estimates authorize the exponential increase of the solution with a rate proportional to the L^∞ norm of σ. Obtaining an explicit energy dissipation result with $\sigma(x)$ variable (which I conjecture to hold) remains to my knowledge an open question.

The general case. Let us come back to the general system (29). The fact that this system is a perturbation of a weakly hyperbolic system remains true : passing from (26) to (29) one doubles (from d to $2d$) the size of the system and doing so, one has added unphysical modes: the associated matrices $A(k)$ have 0 as an eigenvalue with multiplicity d and are not diagonalizable (there is a Jordan block associated with this eigenvalue 0). However, one shows in [6] that, under rather general hypotheses, (29) is weakly well-posed. On the other hand, there is no general stability result and it can not be the case, as we are going to see in the next section.

3.3 On the stability of PML models.

We are going to see in this section that the PML models that we have presented are not always stable. More precisely, we are going to give a condition on the original propagation model which is necessary for the stability of the PML model .This condition is related to the shape of the "wavefronts" in the original system. It is not my purpose to give here the rigorous definition of a wavefront. Roughly speaking, in 2D, the wavefronts are curves which describe the "shape" of a wave emitted from a point source. More rigorously, these curves permit to describe the location of the singularities of this solution, at a given time. Which is important for what follows is that wavefronts can be characterized geometrically in terms of the so-called *slowness curves* which are themselves deduced from the *dispersion relation* of the propagation model.

An example : elastic waves in anisotropic media. We consider now the application of PML's to the propagation of elastic waves in a orthotropic medium.The propagation

equations are given by ($u \in \mathbb{R}^2$ denotes the displacement field and the equations are written in second order form, see remark 17)

$$\rho \frac{\partial^2 u}{\partial t^2} - P\Big(i\frac{\partial}{\partial x}, i\frac{\partial}{\partial y}\Big)\, u = 0, \tag{78}$$

where $\rho > 0$ is the density of the medium and where we have set:

$$P(K_x, K_y) = \begin{pmatrix} c_{11}K_x^2 + c_{33}K_y^2 & (c_{12}+c_{33})K_xK_y \\ (c_{12}+c_{33})K_xK_y & c_{33}K_x^2 + c_{22}K_y^2 \end{pmatrix}. \tag{79}$$

Remark 17 *The system can be put in first order form (26) with $d = 5$ by taking as unknowns the two components of the velocity field (the time) and the three constraints σ_{xx}, σ_{yy} and σ_{xy}. This system is hyperbolic and symmetric under appropriate conditions (see below).*

In order to guarantee the hyperbolicity de (78), the coefficients c_{ij} must satisfy $c_{11} > 0,\ \ c_{22} > 0,\ \ c_{33} > 0,\ \ c_{11}c_{22} - c_{12}^2 > 0$. Isotropic media correspond to :

$$c_{11} = c_{22} = \lambda{+}2\mu,\ c_{12} = \lambda,\ c_{33} = \mu, \qquad \lambda > 0, \quad \mu > 0 \quad (\text{the Lamé coefficients}).$$

The dispersion relation of (78) is:

$$F(\omega, k) \equiv \det\ \big(\, P(k_x, k_y) - \rho\, \omega^2 I\,\big) = 0\,. \tag{80}$$

Note that F is a homogeneous function of degree 4 in (ω, k). For each k, the equation (80) has four solutions of the form $\omega(k) = |k|\ \omega(K), K = k/|k|$, which can be regrouped in two pairs of two opposite real values. The equation (80) can be rewritten as:

$$F\Big(1, \frac{k_x}{\omega}, \frac{k_y}{\omega}\Big) = 0\,. \tag{81}$$

By definition, the *slowness curve* of (78) is the curve of equation (81) in the *slowness* plane $(k_x/\omega, k_y/\omega)$. On obtains in fact the union of two closed curves (that is why we shall speak of two slowness curves), symmetric with respect to the coordinate axes and with respect to the origin. In the isotropic case, these are two circles centered at the origin. Each line passing by the origin, parallel to K, crosses the slowness curves at four points (2 by 2 symmetric with respect to the origin) which are the four $K/\omega(|K|)$'s. The two solutions of smaller modulus correspond to the rapid waves and are called quasi-longitudinal modes (these are the usual longitudinal modes in the isotropic case) and the two solutions of larger modulus correspond to the slow waves and are called quasi-shear modes (these are the usual shear modes in the isotropic case). A rather simple geometrical transformation, that I will not describe here, allows us to pass from one "slowness" curve to the corresponding "wavefront" : one curve is the *polar reciprocal of the other* [5]. We present below the slowness curves and related wave fronts for three particular orthotropic media corresponding to density $\rho = 1$ and

- Medium I : $c_{11} = 4, \quad c_{22} = 20, \quad c_{33} = 2, \quad c_{44} = 3.8$

- Medium II : $c_{11} = 20, \quad c_{22} = 20, \quad c_{33} = 2, \quad c_{44} = 3.8$
- Medium III : $c_{11} = 4, \quad c_{22} = 20, \quad c_{33} = 2, \quad c_{44} = 7.5$

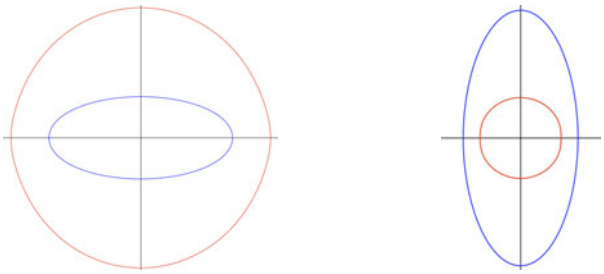


Figure 11: Slowness curves (left) and wavefronts (right) for medium I.

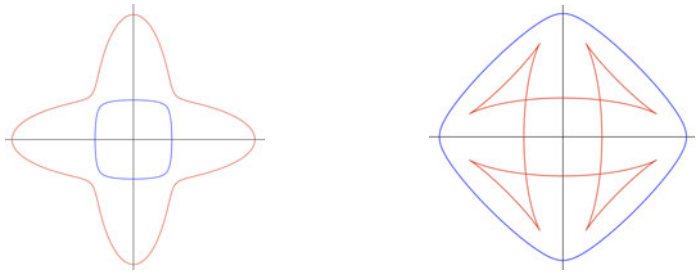


Figure 12: Slowness curves (left) and wavefronts (right) for medium II.

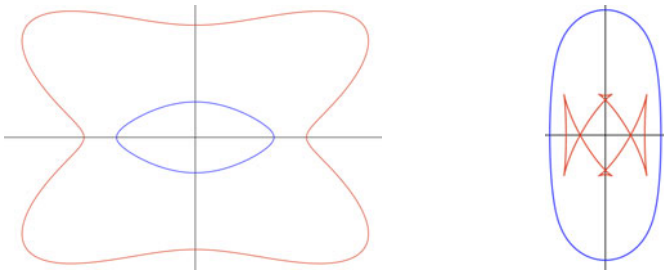


Figure 13: Slowness curves (left) and wavefronts (right) for medium III.

Next, in figures 14 to 17, we present the results of numerical computations corresponding to the propagation of an elastic wave emitted from a point source for four elastic media with density 1 : an isotropic medium with $\lambda = 16$ and $\mu = 2$ and the three above orthotropic media. The physical computational domain (delimited by the dotted line in the pictures) is a square and is surrounded by a PML. The source is located close to the left upper corner of the physical domain. The modulus of the

displacement field is represented at several instants. One observes that everything goes well with the isotropic medium and with the media I and II, while an instability is developing with the medium III. We are going to see that this phenomenon, which is due to the continuous model and not to the numerical scheme that has been used, can be explained by the analysis.

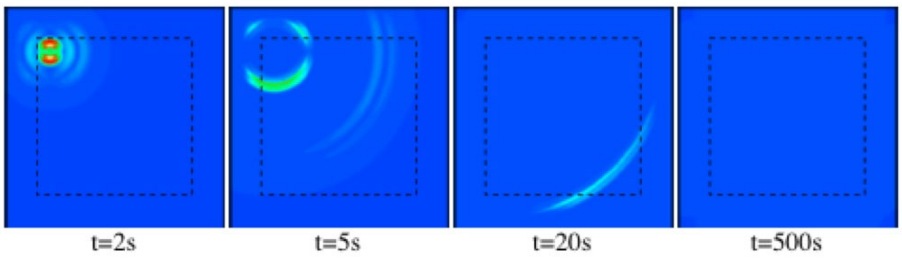


Figure 14: The isotropic medium.

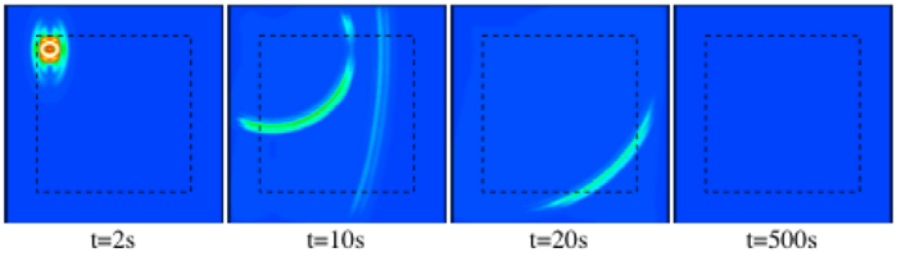


Figure 15: The anisotropic medium I.

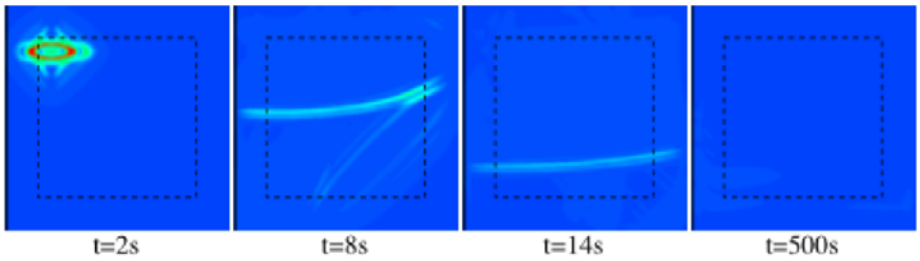


Figure 16: The anisotropic medium II.

A necessary condition for stability. Let us come back to the general system (26). The dispersion relation is written

$$F(\omega, k) := \det\left(\mathbf{A}(k) + \omega I\right) = 0, \quad \mathbf{A}(k) = k_x\, \mathbf{A}_x + k_y\, \mathbf{A}_y. \tag{82}$$

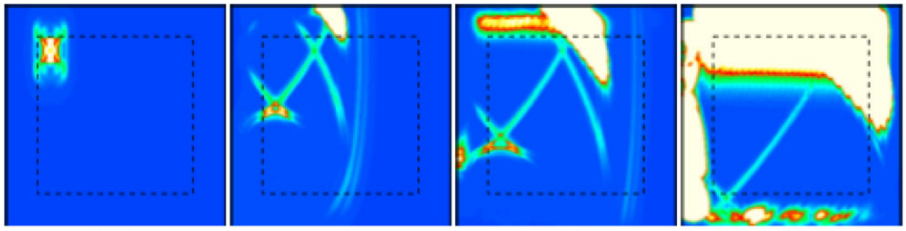


Figure 17: The anisotropic medium III.

where $F(\omega, k)$ is a homogeneous function of degree d. Therefore, the notions of slowness curves introduced in the previous paragraph can easily be generalized to the system (26).

One calls the *group velocity* associated to the mode $\omega(k)$ (one of the solutions of (82), that is a homogeneous function of degree 1) the quantity

$$V_g(k) = \nabla_k \omega(k) \qquad \big(\ = \big(V_x(k), V_y(k)\big) \ \big). \tag{83}$$

It is clearly a homogeneous function of degree 0, that is to say a function of the direction vector $K = k/|k|$ only. Differentiating (82) with respect to k, one sees that one also have the formula

$$V_g(k) = -\Big(\frac{\partial F}{\partial \omega}(\omega, k)\Big)^{-1} \nabla_k F(\omega, k), \quad \big(\text{with } \omega = \omega(k)\big). \tag{84}$$

This formula demonstrates that "the group velocities are orthogonal to the slowness curves". More exactly, the direction of $V_g(k)$ is given by the normal to the slowness curve at the point $k/\omega(k)$. Let us now state the main result of this section:

Theorem 7 *In order that the PML model (in the x-direction) (29) be stable, it is necessary that, for any mode solution of (82),*

$$\forall\ K = (K_x, K_y), \quad K_x \cdot V_x(K) \geq 0. \tag{85}$$

Proof : From the construction of the PML model (see section 2), it is easy to deduce that the dispersion relation of the perturbed system (29) can be obtained by simply replacing in (82) k_x by $k_x \times \big(i\omega/(i\omega + \sigma)\big)$:

$$F\Big(\omega, k_x\big(\frac{i\omega}{i\omega + \sigma}\big), k_y\Big) = 0. \tag{86}$$

Let us introduce the variables

$$c = \frac{\omega}{|k|}, \quad K = \frac{k}{|k|}, \quad \varepsilon = \frac{\sigma}{|k|}. \tag{87}$$

The relation (86) can be rewritten, thanks to homogeneity,

$$F\Big(c, K_x\big(1 - i\,\frac{\varepsilon}{c}\big), K_y\Big) = 0, \tag{88}$$

which we consider as an equation in c. For $\varepsilon = 0$, (i. e. $\sigma = 0$ ou $|k| \to +\infty$) we obtain as particular real solutions $c = c(K) := \omega(K)$ ($\equiv \omega(k)/|k|$), namely the phase velocities of the "physical modes" (let us remind that the fact that $\omega(k)$ is real is the consequence of the hyperbolicity of (26)). Let us fix the direction K. By the implicit function theorem, it is easy to show that, in the neighborhood of each $c(K)$, the equation (88) has a branch of solutions $\varepsilon \to c(K, \varepsilon)$ which admits the following Taylor expansion for small ε

$$c(K, \varepsilon) = c(K) + \alpha(K)\, \varepsilon + O(\varepsilon^2).$$

Substituting this equality into (88), we obtain

$$F\big(c(K), K\big) + \varepsilon\, \alpha(K)\, \frac{\partial F}{\partial \omega}\big(c(K), K\big) + i\varepsilon\, \frac{K_x}{c(K)} \frac{\partial F}{\partial k_x}\big(c(K), K\big) + O(\varepsilon^2) = 0,$$

from which we deduce, taking into account (84) (with $\omega = \omega(k)$)),

$$\alpha(K) = -\, i\, \left(\frac{\partial F}{\partial \omega}(\omega, k)\right)^{-1} \left(\frac{K_x}{c(K)} \frac{\partial F}{\partial K_x}(\omega, k)\right) = i\, \frac{K_x \cdot V_x(K)}{c(K)}\ . \tag{89}$$

Going back to original variables (ω, k) and σ, we have shown that for k large enough, the equation in ω (86) admits, "at the neighborhood of each physical mode $\omega(k)$", a branch of solution $k \to \omega(k, \sigma)$ such that

$$\omega(k, \sigma) = \omega(k) + i\, \frac{K_x \cdot V_x(K)}{c(K)}\, \sigma + O\Big(\frac{1}{|k|}\Big), \quad (|k| \to +\infty). \tag{90}$$

One then concludes easily: if $K_x \cdot V_x(K) < 0$, for sufficient large $|k|$, $\omega(k, \sigma)$ belongs to the lower complex half plane, which contradicts stability (cf. (60)) . ■

Remark 18 *Obviously,to obtain a necessary condition for the PML model in the y direction, it suffices to replace V_x by V_y in (85). Moreover, the condition (85) might be non sufficient. We refer for instance to [6] for a more complete study in the case of elastodynamics equations.*

The property (85) can be interpreted geometrically in the following way. If one draws the vector that joins (in the $K = (K_x, K_y)$-plane) the origin to any point $S(K) = K/c(K)$ of the slowness curve and if from this point one draws a second vector which has the same direction than the corresponding group velocity vector, the resulting " broken vector" must cross the vertical line (I mean the line parallel to the K_y direction) passing by the point $S(K)$ (cf. figure (18)). In the contrary, one says that the model admits "backward propagative modes" (or "inverse modes"). Thus, the condition (85) corresponds to the absence of "backward propagative modes".

With this geometrical interpretation, it is easy to " explain" the numerical results of figures 14 to 17. First, unstable modes as the one of figure (85) can not exist when the slowness curve "is convex" (more precisely when the bounded domain delimited by this curve is convex) which is the case for an isotropic medium or for the medium II.

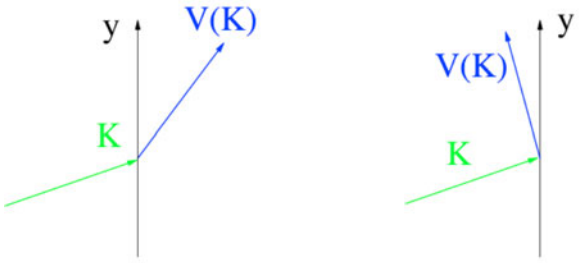


Figure 18: Left : stable mode. Right : unstable mode

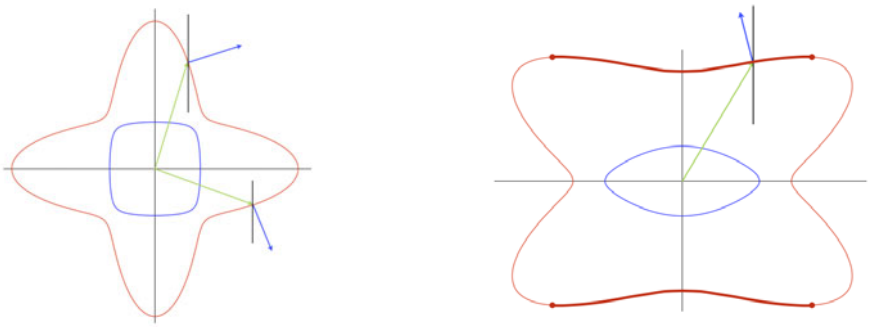


Figure 19: Left : medium II. Right : medium III

Next, to understand the difference between the media II and III (for which the above convexity property is lost for the quasi-shear waves), it suffices to have a look to the figure 19. Contrary to the medium II, the medium III admits unstable modes that correspond to the thicker part of the slowness curve. One also understands easily the the system of elastodynamics is not the only one, among physically relevant wave propagation models, for which the PML model (29) is unstable. This is for instance the case of the anisotropic wave equation

$$\frac{\partial^2 u}{\partial t^2} - \text{div}\,(A\nabla u) = 0, \quad A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad a\,c - b^2 > 0, \quad b \neq 0. \tag{91}$$

or linearized Euler equations (one linearizes Euler equations in a compressible fluid around a particular solution corresponding to a uniform stationary flow propagating in the x direction with Mach number M - the equations are expressed in a coordinate

system such that the speed of sound in the same fluid at rest):

$$\begin{cases} \dfrac{\partial u}{\partial t} + M\,\dfrac{\partial u}{\partial x} - \dfrac{\partial v_x}{\partial x} - \dfrac{\partial v_y}{\partial y} = 0, \\ \dfrac{\partial v_x}{\partial t} + M\,\dfrac{\partial v_x}{\partial x} - \dfrac{\partial u}{\partial x} = 0, \\ \dfrac{\partial v_y}{\partial t} + M\,\dfrac{\partial v_y}{\partial x} - \dfrac{\partial u}{\partial y} = 0\ . \end{cases} \tag{92}$$

We visualize in the figure 20 the slowness curves and the back propagative modes in each case.

Of course, in the unstable situations, one needs to construct a new PML model, really different from the model (29) of Bérenger's type in the sense that it would not be only a reformulation "à la Zhao-Cangellaris" (see section 3). As a matter of fact, satisfactory solution have been proposed for the anisotropic wave equation (91) (see [17]) or for the linearized Euler equations (see [2], [22] or [13]). The case of the anisotropic elastic waves remains, to my knowledge a challenging open problem.

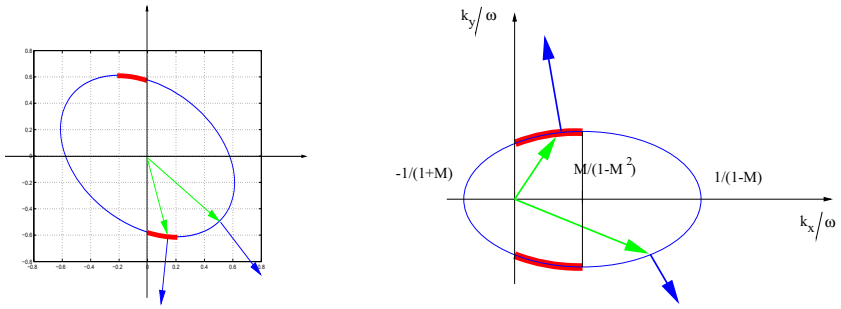


Figure 20: Slowness curves and unstable modes for (91) (left) and (92) (right).

About error estimates. Much less has been done in this direction for time dependent problem. Let us mention however [13], where one has been able to compute explicitly the Green's function of the 2D acoustic wave equation with a finite PML, even in the case of a variable profile σ. This computation is based on the Cagniard-De Hoop method and exploits in some sense the construction of the PML model via a complex change of variable. As a by product, this computation shows in this case the stability of the PML model even in the case of a variable σ. Using this Green's function, one can derive sharp error estimates which constitute a 2D equivalent of theorem 2 and show that PML are "exponentionnally accurate". These estimates also emphasize the role of the time in the error estimates, as well as the distance between the support of the data of the problem and the artificial boundary.

4 Conclusion

In this text, we have given an elementary introduction to the technique of Perfectly Matched Layers for realizing almost transparent boundary conditions for the numerical simulation of linear wave propagation in unbounded media. Various aspects, from the construction of the method to the theory via more practical and numerical aspects have been overviewed and various open questions have been mentioned : the method of Perfectly Matched Layers s still an active field of research for applied mathematicians.

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